CSE 270 Sports Analytics

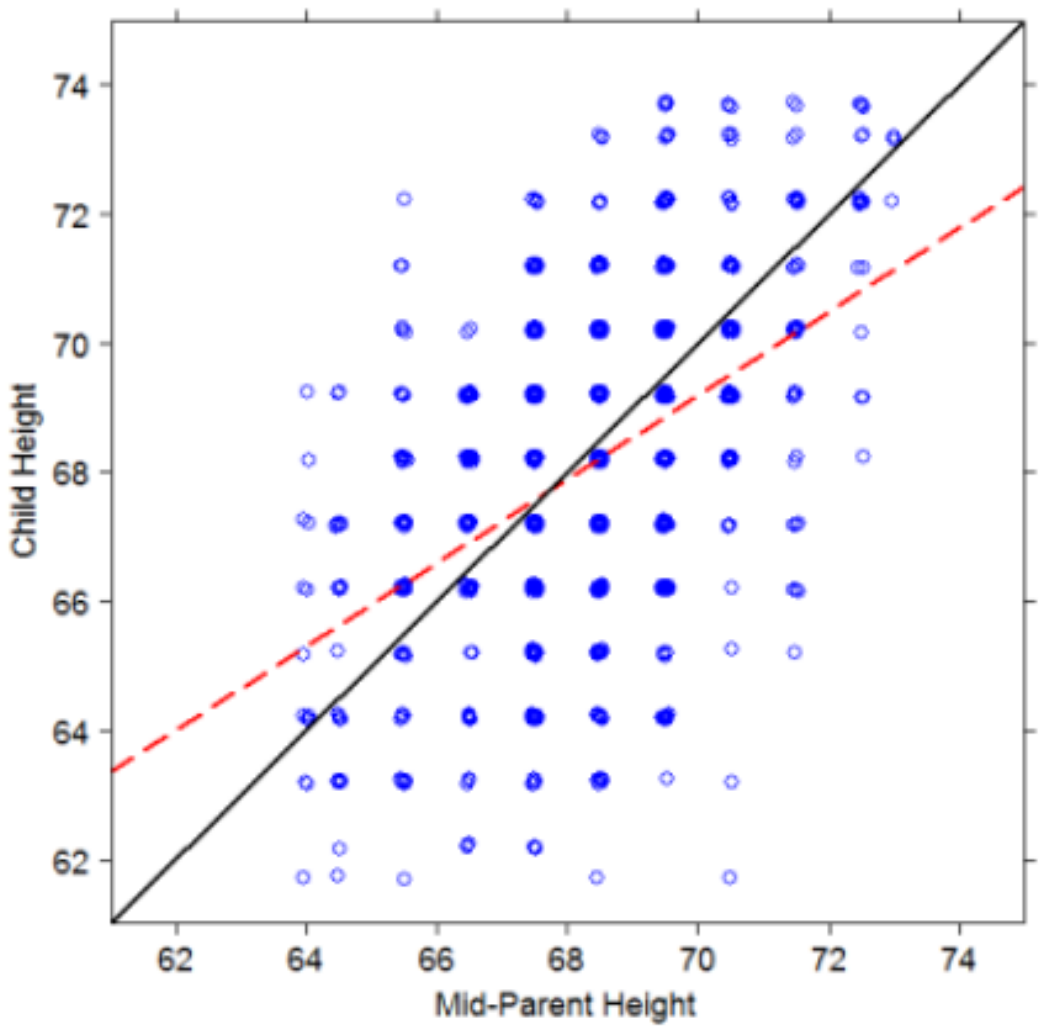
Regresion to the mean

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July 14, 2018

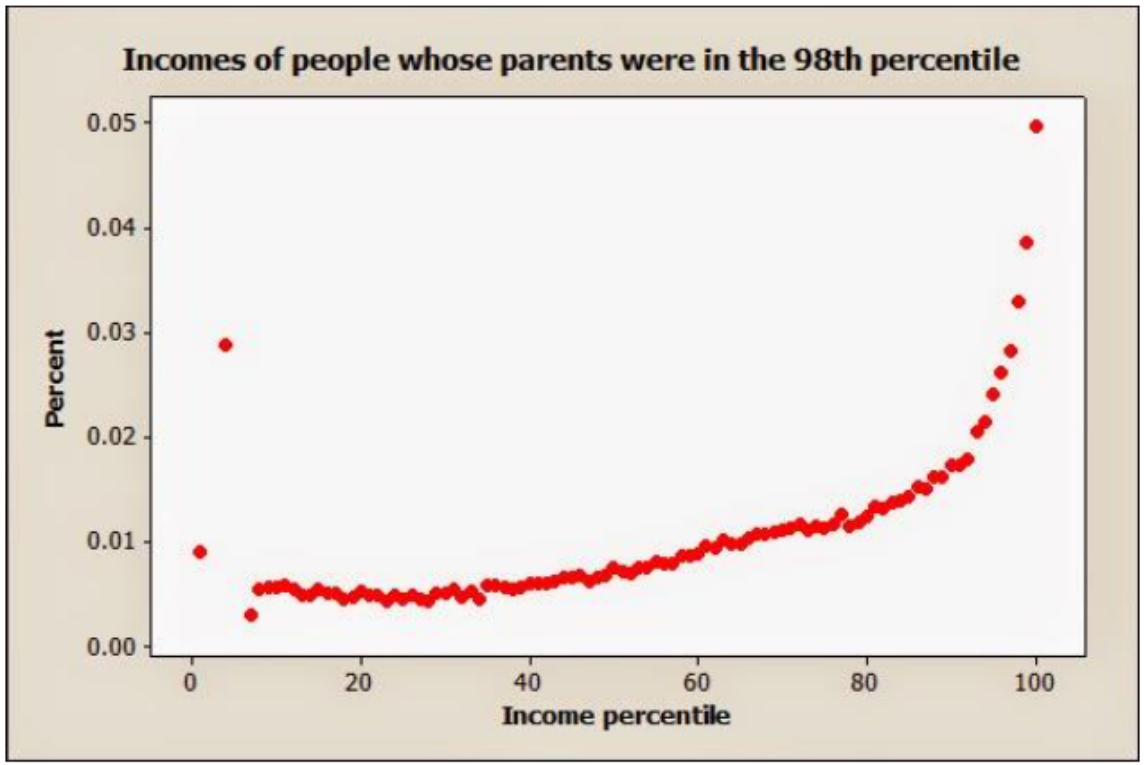
Regression to the mean: Height

Sir Francis Galton proved that the kids of shorter parents have higher height than their parents, and kids of taller parents have shorter height than their parents



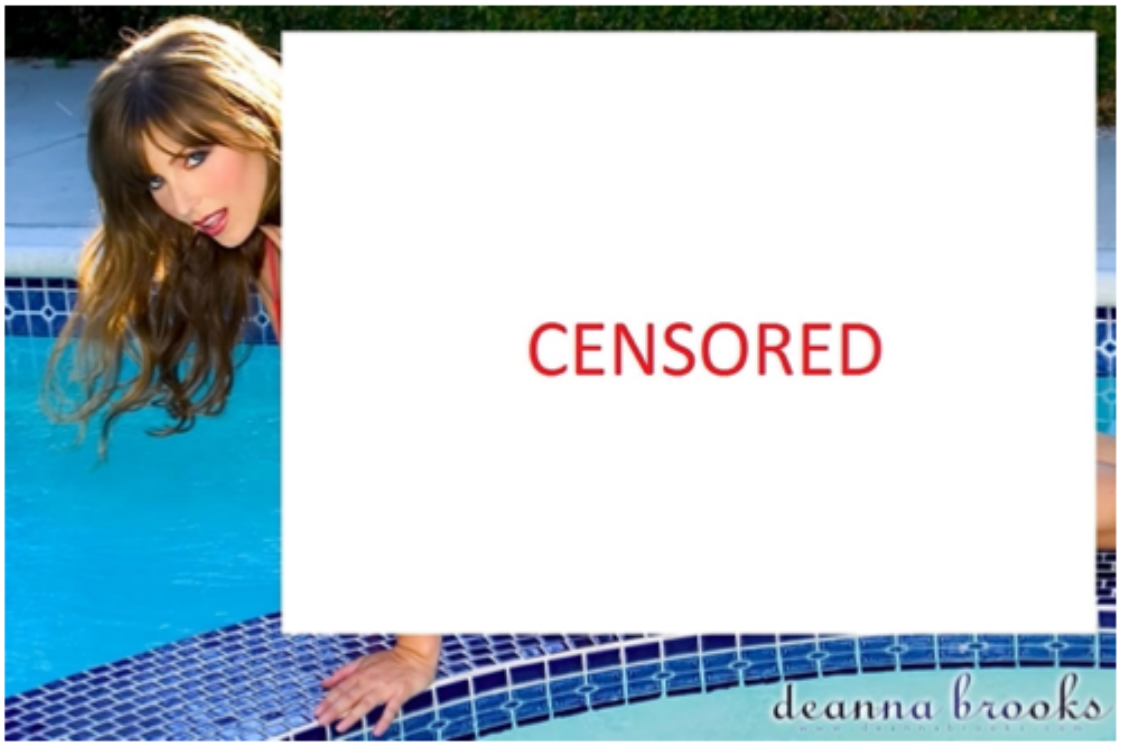
Regression to the mean: Wealth

If your income is at the 98th percentile of the income distribution — that is, you earn more than 98 percent of the population — the best guess is that your children, when they are adults, will be in the 65th percentile.

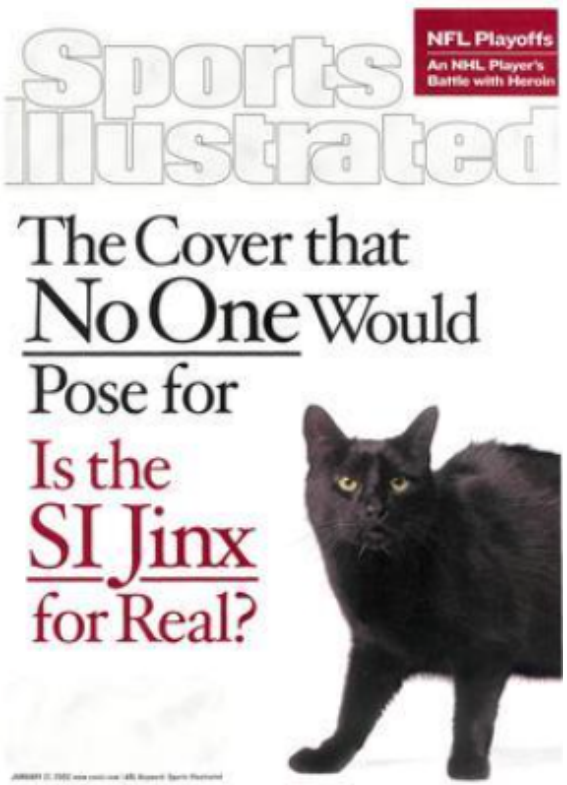


Regression to the mean: Examples

IN 2006, TRADINGMARKETS, a company that helps people trade stocks, asked ten Playboy Playmates to select five stocks each. The idea was to see if they could beat the market. The winner was Deanna Brooks, Playmate of the Month in May 1998. The stocks she picked rose 43.4 percent, trouncing the S& P 500, which gained 13.6 percent, and beating more than 90 percent of the money managers who actively try to outperform a given index. Brooks wasn't the only one who fared well. Four of the other ten Playmates had better returns than the S& P 500 while less than a third of the active money managers did.

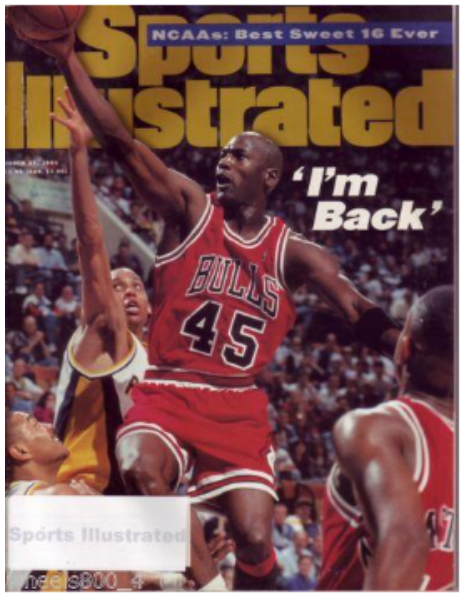


Sports Illustrated cover Jynx



***The Sports Illustrated* cover jinx** is an urban legend that states that individuals or teams who appear on the cover of the Sports Illustrated magazine will subsequently be jinxed (experience bad luck)

Exception



::::

Regression to the mean: Definitions

Regression to the mean is a statistical phenomenon that can make natural variation in repeated data look like real change. It happens when unusually large or small measurements tend to be followed by measurements that are closer to the mean.

Regression to the mean is a technical way of saying that things tend to even out over time.

The term [**Regression to the mean**] is now generally used to label the phenomenon that a variable that is extreme on its first measurement will tend to be closer to the center of the distribution for a later measurement.

True score theory

From the perspective of classical test theory, an examinee's obtained test score (X) is composed of two components, a true score component (T) and an error component (E)

$$X = T + E$$

Skills and Luck: the formula

$$Outcome = Skills + Luck$$

$$Skills = Outcome - Luck$$

How much is the skill and how much is the luck ?

$$Skills = Outcome - Luck$$

$$Var(Skills) = Var(Outcome) - Var(luck)$$

Example: NBA

We will assume that the game outcome has Binomial distribution

$$P(\text{win x games}) = \frac{n!}{x!(n-x)!} p^x * (1-p)^{(n-x)}$$

$$\mu = np$$

$$\sigma^2 = np(1-p)$$

Example: NBA

```
library(SportsAnalytics270)
data("nba_misc")
head(nba_misc)
```

```
##   Rk                  Team  Age  W  L PW PL  MOV   SOS  SRS  ORtg  DRtg  Pace
## 1  1        Milwaukee Bucks 26.9 60 22 61 21 8.87 -0.82 8.04 113.8 105.2 103.3
## 2  2  Golden State Warriors 28.4 57 25 56 26 6.46 -0.04 6.42 115.9 109.5 100.9
## 3  3        Toronto Raptors 27.3 58 24 56 26 6.09 -0.60 5.49 113.1 107.1 100.2
## 4  4             Utah Jazz 27.3 50 32 54 28 5.26  0.03 5.28 110.9 105.7 100.3
## 5  5        Houston Rockets 29.2 53 29 53 29 4.77  0.19 4.96 115.5 110.7  97.9
## 6  6 Portland Trail Blazers 26.2 53 29 51 31 4.20  0.24 4.43 114.7 110.5  99.1
##     FTr X3PAr   TSP eFGP_o TOVP_o ORBP_o FT_FGA_o eFGP_d TOVP_d DRBP_d FT_FGA_d
## 1 0.255 0.419 0.583  0.550   12.0   20.8    0.197  0.503   11.5   80.3    0.162
## 2 0.227 0.384 0.596  0.565   12.6   22.5    0.182  0.508   11.7   77.1    0.205
## 3 0.247 0.379 0.579  0.543   12.4   21.9    0.198  0.509   13.1   77.1    0.190
## 4 0.295 0.394 0.572  0.538   13.4   22.9    0.217  0.507   12.4   80.3    0.189
## 5 0.279 0.519 0.581  0.542   12.0   22.8    0.221  0.525   13.4   74.4    0.210
## 6 0.258 0.339 0.568  0.528   12.1   26.6    0.210  0.516   11.0   77.9    0.195
##                     Arena Attendance
## 1            Fiserv Forum     721692
## 2            Oracle Arena     803436
## 3        Scotiabank Arena     812822
## 4 Vivint Smart Home Arena     750546
## 5           Toyota Center     740392
## 6             Moda Center     799345
```

Example: NBA

Every team has played 82 games. Create another variable with win ratio

```
nba_misc$WR<-nba_misc$W/82
```

Lets get the variance of the winning ratio- This is the variance for outcome

```
var(nba_misc$WR)
```

```
## [1] 0.02151839
```

Let us look at the variance for the luck. Here we assume that the chance of winning for each team is the same 0.5.

```
0.5*0.5/82
```

```
## [1] 0.00304878
```

For variance of the luck (proportion p), we use the following formula

$$\frac{p(1-p)}{n}$$

Example: NBA

Var(Skills) = Var(Outcome) - Var(Luck)

Var(Skills) = 0.02151839 - 0.00304878

Var(Skills) = 0.01846961

The paradox fo Skill:
More skills means luck is more important

Assuming that the variance in luck stays constant

Example: NBA

Var(Skills) = Var(Outcome) - Var(Luck)

Var(Skills) = 0.01861577 - 0.00304878

Var(Skills) = 0.01556699

This analysis allows us to look at the ratio of variance (luck) to variance (observed) in order to determine the contribution of luck.

$$\frac{0.00304878}{0.01556699} = 19\%$$

Paradox of skill

The less variation in skill there is, the more important luck is.

Skill or luck ?

Manchester United vs Bayern Munich UCL Final

1999

https://youtu.be/YnecJa5satA?t=112

Manchester United vs Bayern Munich, UCL final 1999

Overall
Box-Score Statistics

Statistic	MU	Bayern
Goals Scored	2	1
Total Shots	15	15
Shots on target	9	7
Ball possession	53%	47%
Corner kicks	12	7
Fouls committed	11	10
Offsides	6	8
Yellow cards	0	1
Red cards	0	0

So what ?

The contribution of luck in some professional sports leagues

League	Contribution of luck	Five seasons ended
National Basketball Association	12%	2011
Premier League	31%	2011
Major League Baseball	34%	2011
National Football League	38%	2011
National Hockey League	53%	2012

Source: Analysis by author.

Mauboussin, Michael J.. The Success Equation: Untangling Skill and Luck in Business, Sports, and Investing. Harvard Business Review Press.

Curse of the Bambino



- In 1919 Red Sox sold their star player, Babe Ruth to New York Yankees
- Red Sox was not able to win World Series for the next 86 years, despite of being a favorite in many final games

Regression to the mean

Construct the variable DF of difference

```
data(nba_east)
data(nba_west)
nba_st <- rbind(nba_east,nba_west)


nba_st$PW <- nba_st$PS^14.105/(nba_st$PS^14.105+nba_st$PA^14.105)
nba_st$G <- nba_st$W+nba_st$L
nba_st$PW_G <- round(nba_st$PW*nba_st$G)

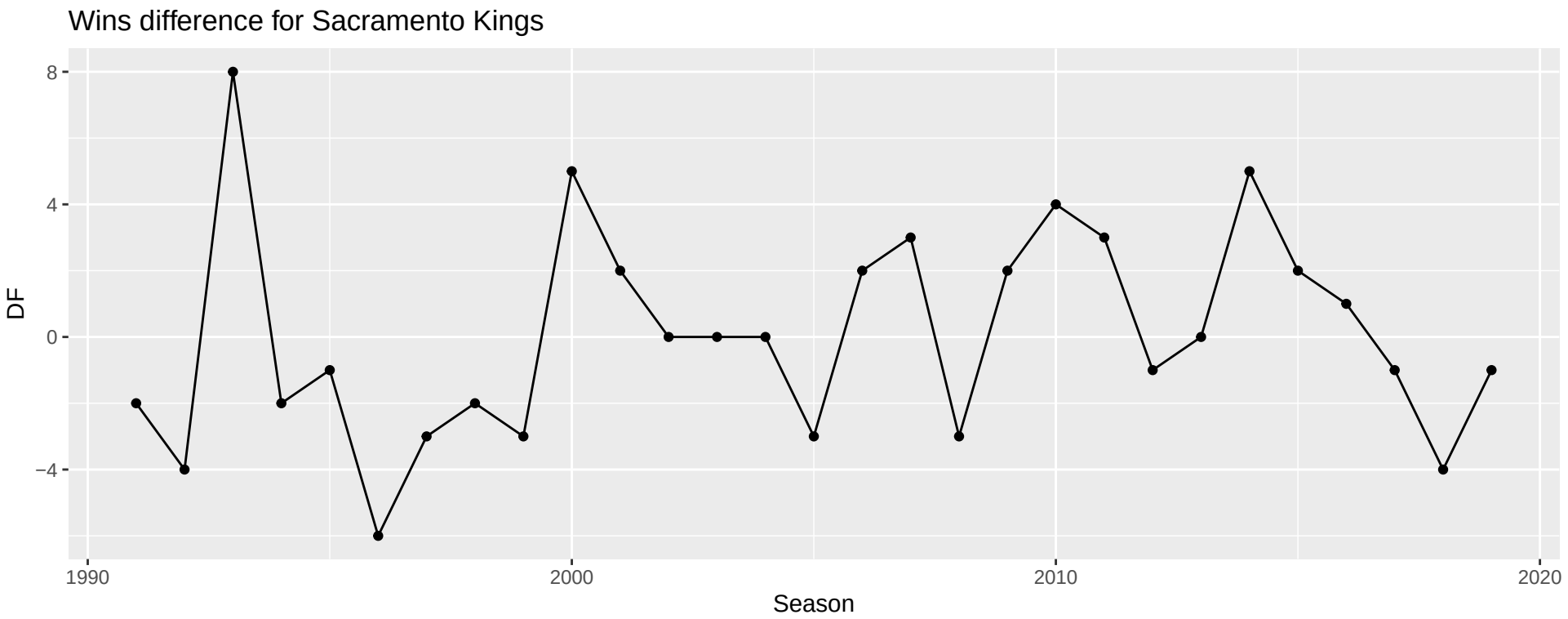
nba_st$DF <- nba_st$PW_G-nba_st$W
```

Regression to the mean

Let's plot for one team

```
nba_st$DF <- nba_st$PW_G-nba_st$W

nba_st %>% filter(Team=="Sacramento Kings") %>%
ggplot(aes(x=Season, y=DF))+geom_point()+geom_line()+
  ggtitle('Wins difference for Sacramento Kings')
```



Regression to the mean

```
means <- nba_st %>% group_by(Team) %>% summarise(mean=mean(DF))
head(means)
```

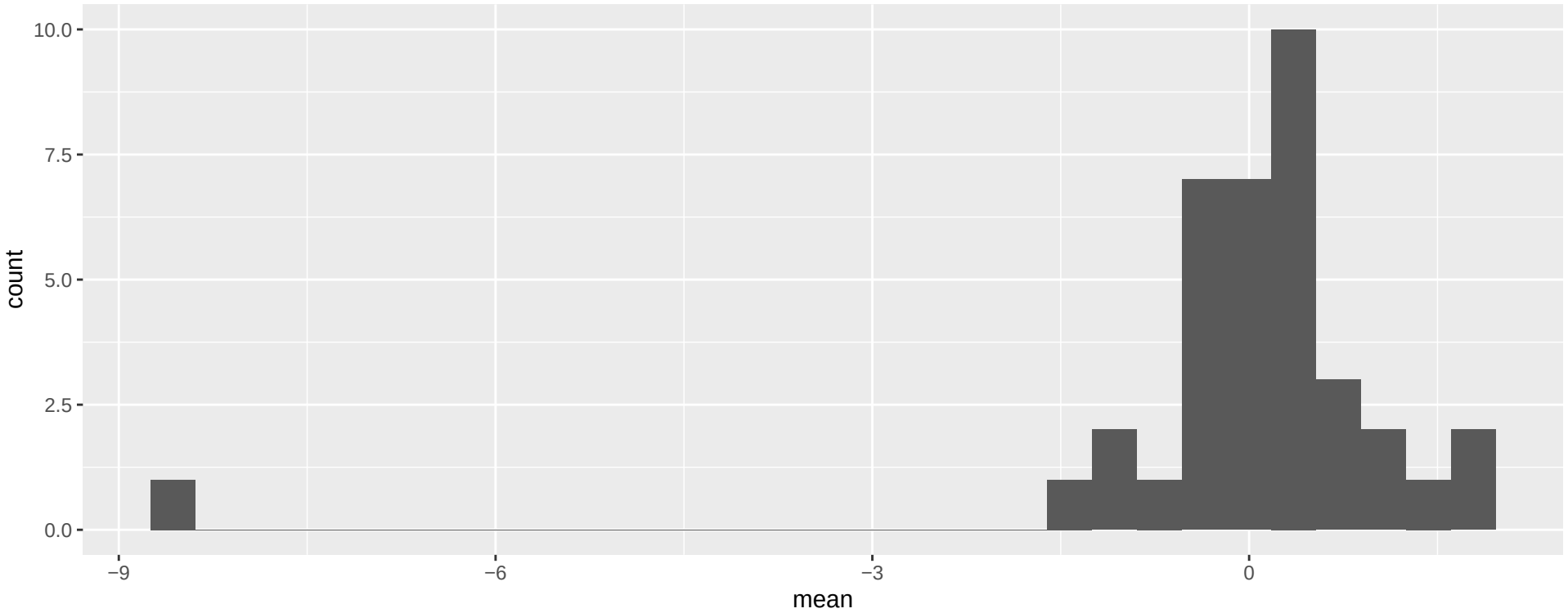
```
## # A tibble: 6 x 2
##   Team                 mean
##   <chr>               <dbl>
## 1 Atlanta Hawks      -0.655
## 2 Boston Celtics      0.310
## 3 Brooklyn Nets      -1
## 4 Charlotte Bobcats  -0.5
## 5 Charlotte Hornets  -0.353
## 6 Chicago Bulls      -0.897
```

Average difference by teams (You need dplyr here)

Regression to the mean

The histogram of the averages

```
ggplot(means, aes(x=mean))+geom_histogram()
```



Competitive Balance in Sports

Competitive Balance in Sports

Competitive Balance

Definition from economics:

A market situation where no business is too big or has an unfair advantage. Source: http://lexicon.ft.com/Term?term=competitive-balance

Definition in Sports:

The degree of uncertainty in the match outcome or the final league standings.

Higher is the Competitive Balance, higher is the uncertainty

Competitive Balance: Dimensions

Long-run

Distribution of championships over time

Seasonal

Within season

Short-run

Match uncertainty Outcome of contest

Measuring Seasonal Dimensions

- Statistical measures
  - Range, variance or deviation
- Standard deviation (SD)
  - Dispersion from mean
  - If SD is low, then CB is high
- CB is often measured by
  - SD of actual wins $\div$ idealized SD
  - As measure increases, competitive balance worsens

Seriea: Measuring balance with standard deviation

```
library(SportsAnalytics270)
data("seriea_st")
colnames(seriea_st)
```

```
##  [1] "Rank"   "Team"   "M"      "W"      "D"      "L"      "Goals"  "Diff"
##  [9] "Pt"     "Season"
```

The dataset

```
head(seriea_st)
```

```
##   Rank           Team  M  W  D L Goals Diff Pt Season
## 1    1       AC Milan 34 21 10 3 60:24   36 73   1996
## 2    2       Juventus 34 19  8 7 58:35   23 65   1996
## 3    3     Lazio Roma 34 17  8 9 66:38   28 59   1996
## 4    4 ACF Fiorentina 34 17  8 9 53:41   12 59   1996
## 5    5        AS Roma 34 16 10 8 51:34   17 58   1996
## 6    6       Parma AC 34 16 10 8 44:31   13 58   1996
```

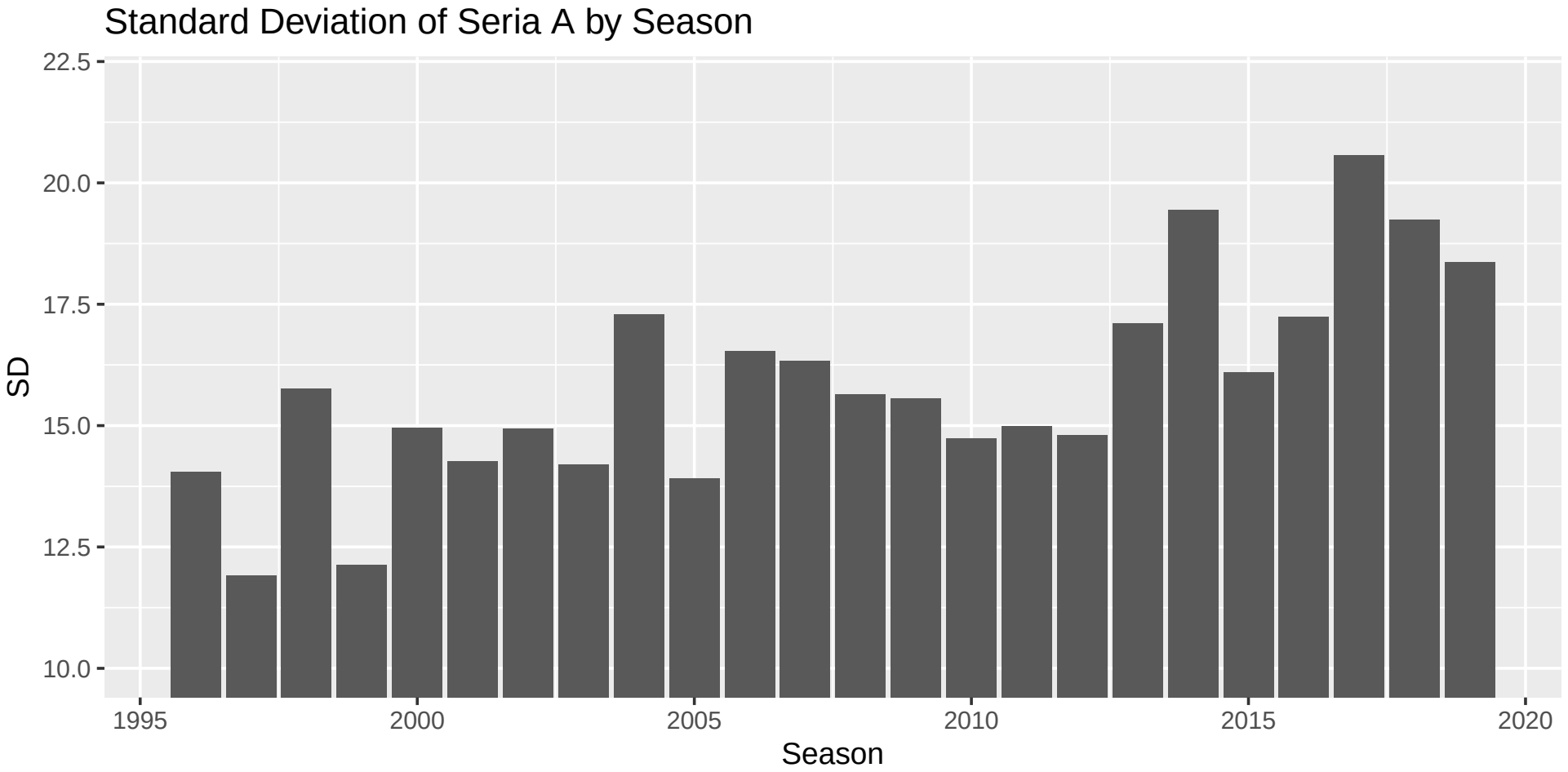
Standard Deviations: Serie A

```
library(dplyr)
seriea_s<-seriea_st %>%
  group_by(Season) %>%
  summarise(SD=sd(Pt))
seriea_s
```

```
## # A tibble: 24 x 2
##    Season    SD
##     <dbl> <dbl>
##  1   1996  14.0
##  2   1997  11.9
##  3   1998  15.8
##  4   1999  12.1
##  5   2000  15.0
##  6   2001  14.3
##  7   2002  14.9
##  8   2003  14.2
##  9   2004  17.3
## 10   2005  13.9
## # ... with 14 more rows
```

Standard Deviations: Serie A

```
ggplot(seriea_s, aes(Season, SD))+geom_bar(stat="identity")+
  ggtitle("Standard Deviation of Seria A by Season")+
  coord_cartesian(ylim=c(10,22))
```



NFL

NFL data

```
data(nfl_st)
summary(nfl_st)
```

```
##       Tm                  W                L                T
##  Length:667         Min.   : 0.000   Min.   : 0.000   Min.   :0.00000
##  Class :character   1st Qu.: 6.000   1st Qu.: 6.000   1st Qu.:0.00000
##  Mode  :character   Median : 8.000   Median : 8.000   Median :0.00000
##                     Mean   : 7.987   Mean   : 7.987   Mean   :0.02699
##                     3rd Qu.:10.000   3rd Qu.:10.000   3rd Qu.:0.00000
##                     Max.   :16.000   Max.   :16.000   Max.   :1.00000
##       Wpct              PF              PA              PD
##  Min.   :0.0000   Min.   :161.0   Min.   :165.0   Min.   :-261.0
##  1st Qu.:0.3750   1st Qu.:298.0   1st Qu.:309.0   1st Qu.: -73.5
##  Median :0.5000   Median :345.0   Median :344.0   Median :   2.0
##  Mean   :0.5002   Mean   :348.3   Mean   :348.3   Mean   :   0.0
##  3rd Qu.:0.6250   3rd Qu.:395.0   3rd Qu.:388.0   3rd Qu.:  72.0
##  Max.   :1.0000   Max.   :606.0   Max.   :517.0   Max.   : 315.0
##       MoV                 SoS                SRS
##  Min.   :-16.300000   Min.   :-5.900000   Min.   :-17.40000
##  1st Qu.: -4.600000   1st Qu.:-1.200000   1st Qu.: -4.35000
##  Median :  0.100000   Median : 0.000000   Median :  0.00000
##  Mean   : -0.001799   Mean   : 0.001049   Mean   :  0.00015
##  3rd Qu.:  4.500000   3rd Qu.: 1.300000   3rd Qu.:  4.50000
##  Max.   : 19.700000   Max.   : 4.300000   Max.   : 20.10000
##       OSRS               DSRS              Season      Conf          Div
##  Min.   :-12.3000   Min.   :-9.800000   Min.   :1998   AFC:335   Central: 43
##  1st Qu.: -3.2000   1st Qu.:-2.300000   1st Qu.:2003   NFC:332   East   :176
##  Median : -0.1000   Median : 0.100000   Median :2008             North  :136
##  Mean   : -0.0006   Mean   :-0.001199   Mean   :2008             South  :136
##  3rd Qu.:  2.7000   3rd Qu.: 2.500000   3rd Qu.:2013             West   :176
##  Max.   : 15.9000   Max.   : 9.800000   Max.   :2018
```

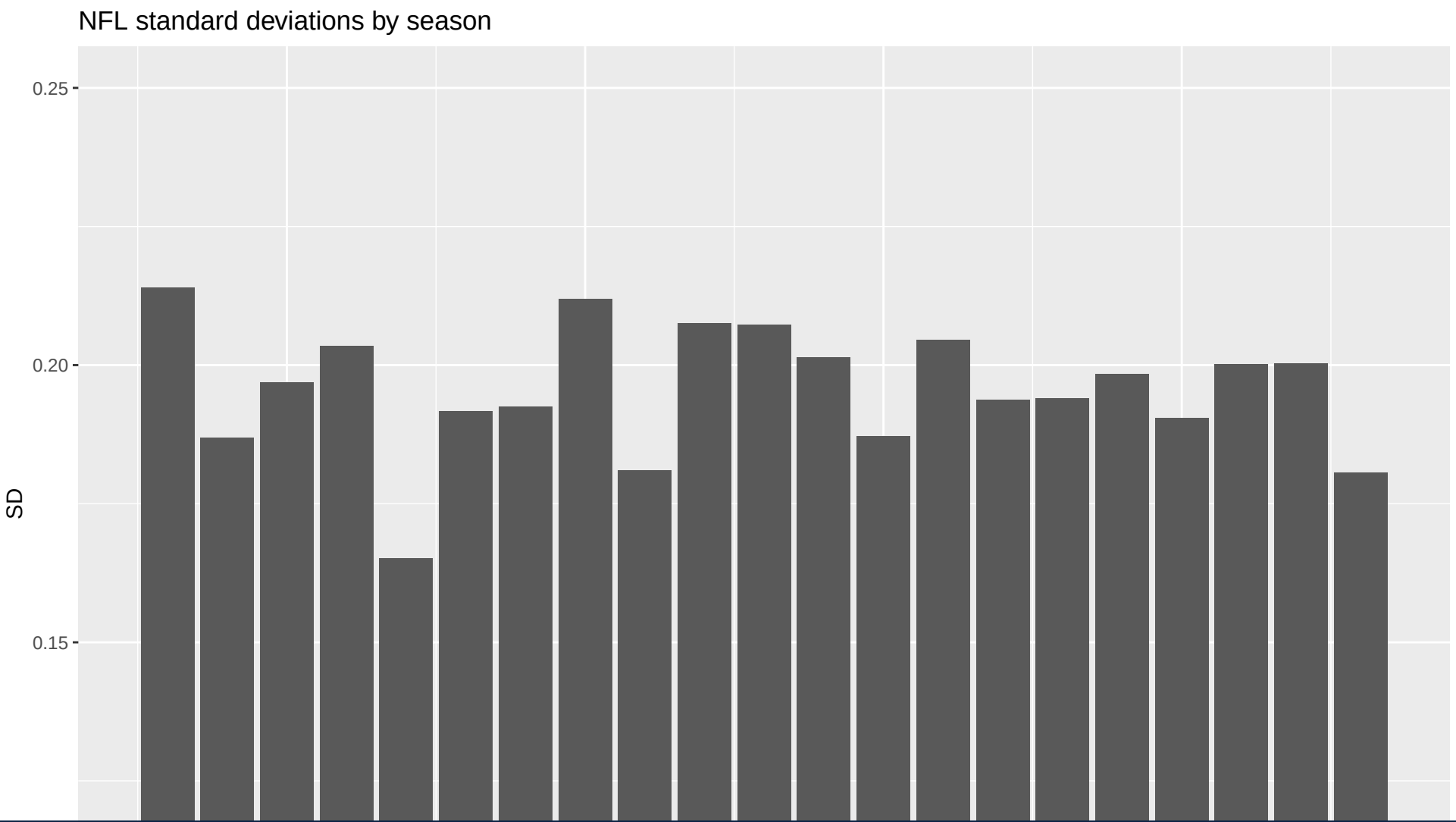
NFL data: SD

```
library(dplyr)
nfl_s<-nfl_st %>%
  group_by(Season) %>%
  summarise(SD=sd(Wpct))
nfl_s
```

```
## # A tibble: 21 x 2
##    Season    SD
##     <int> <dbl>
##  1   1998 0.214
##  2   1999 0.187
##  3   2000 0.197
##  4   2001 0.203
##  5   2002 0.165
##  6   2003 0.192
##  7   2004 0.192
##  8   2005 0.212
##  9   2006 0.181
## 10   2007 0.208
## # ... with 11 more rows
```

NFL data: SD

```
ggplot(nfl_s, aes(Season, SD))+geom_bar(stat="identity")+
  ggtitle("NFL standard deviations by season")+
  coord_cartesian(ylim=c(0.1,0.25))
```



StDev approach: the drawbacks

- Depends on how the game is played and how the points are distributed
- Cannot be used to compare different sports

Measuring Seasonal Dimensions

There are several drawbacks of using standard deviation as a measure of competitive balance.

n and the number of the games played are not always constant over time or same for different sports and leagues.

To deal with this problem, σ is benchmarked against standard deviation that describes dispersion for the league with ideal competitive balance.

Measuring Seasonal Dimensions

Lets assume we are dealing with a game, where the match outcome is either win or loose.

As previously, we will define w (wins ratio) as a number of games a team has won, divided by the number of games it has played.

The standard deviation of the wins ratio will be defined as

$$\sigma = \sqrt{\frac{\sum_{i=1}^{n} (w_i - 0.5)^2}{n}}$$

n - number of teams

w_i - winning ration of team i

Measuring Seasonal Dimensions

Ideally, each team will have 0.5 probability of winning any game (ignoring home-field advantage). Then the ideal standard deviation will be calculated with the following formula

$$\sqrt{\frac{0.5*(1-0.5)}{m}} = \frac{0.5}{\sqrt{m}}$$

Where m is the number of games played by each team.

This is also called Noll-Scully number

Measuring Seasonal Dimensions

Now we can define relative measure of competitive balance

$$\sigma_{\text{relative}} = \frac{\sigma}{0.5/\sqrt{m}}$$

This is also called Noll-Scully number

Measuring Seasonal Dimensions: Noll-Scully number

- Ideally, Noll-Scully number will be equal to 1
- Higher is the number, lower is Competitive Balance
- Higher is the number, lower is Competitive Balance
- Closer is the number to 1, higher is Competitive Balance

Measuring Seasonal Dimensions: Noll-Scully number

The Noll-Scully compares what a league actually looks like to what it would ideally look like. In a perfect world — i.e. one where actual balance matches the ideal — the metric would return a 1.0, meaning the league is balanced.

If the number is above 1.0 it means the teams are further away in wins than we would expect given the ideal.

If the number is below 1.0 — which is very rare case— it means the teams are closer in wins than we would expect.

Noll-Scully number: Calculations

Noll-Scully number

```
#Ideal StDev
id_s<-0.5/sqrt(16)
# Win percentage StDev
sd_wpct<-sd(nfl_st$Wpct)
# Noll-Scully number
sd_wpct/id_s
```

```
## [1] 1.543827
```

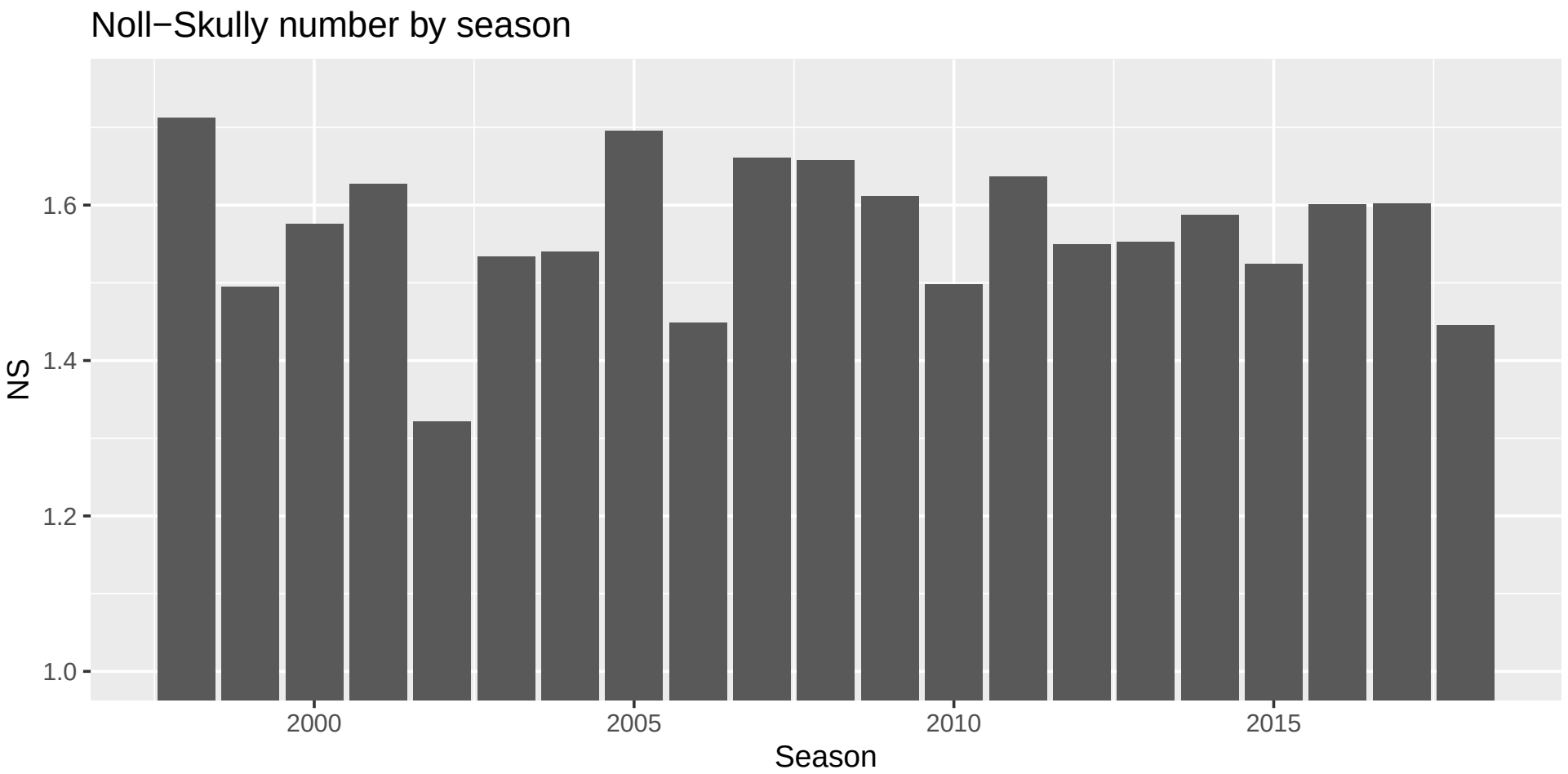
Calculating season-by season

```
library(dplyr)
nfl_s<-nfl_st %>%
  group_by(Season) %>%
  summarise(NS=sd(Wpct)/id_s)
nfl_s
```

```
## # A tibble: 21 x 2
##    Season    NS
##     <int> <dbl>
##  1   1998  1.71
##  2   1999  1.49
##  3   2000  1.58
##  4   2001  1.63
##  5   2002  1.32
##  6   2003  1.53
##  7   2004  1.54
##  8   2005  1.70
##  9   2006  1.45
## 10   2007  1.66
## # ... with 11 more rows
```

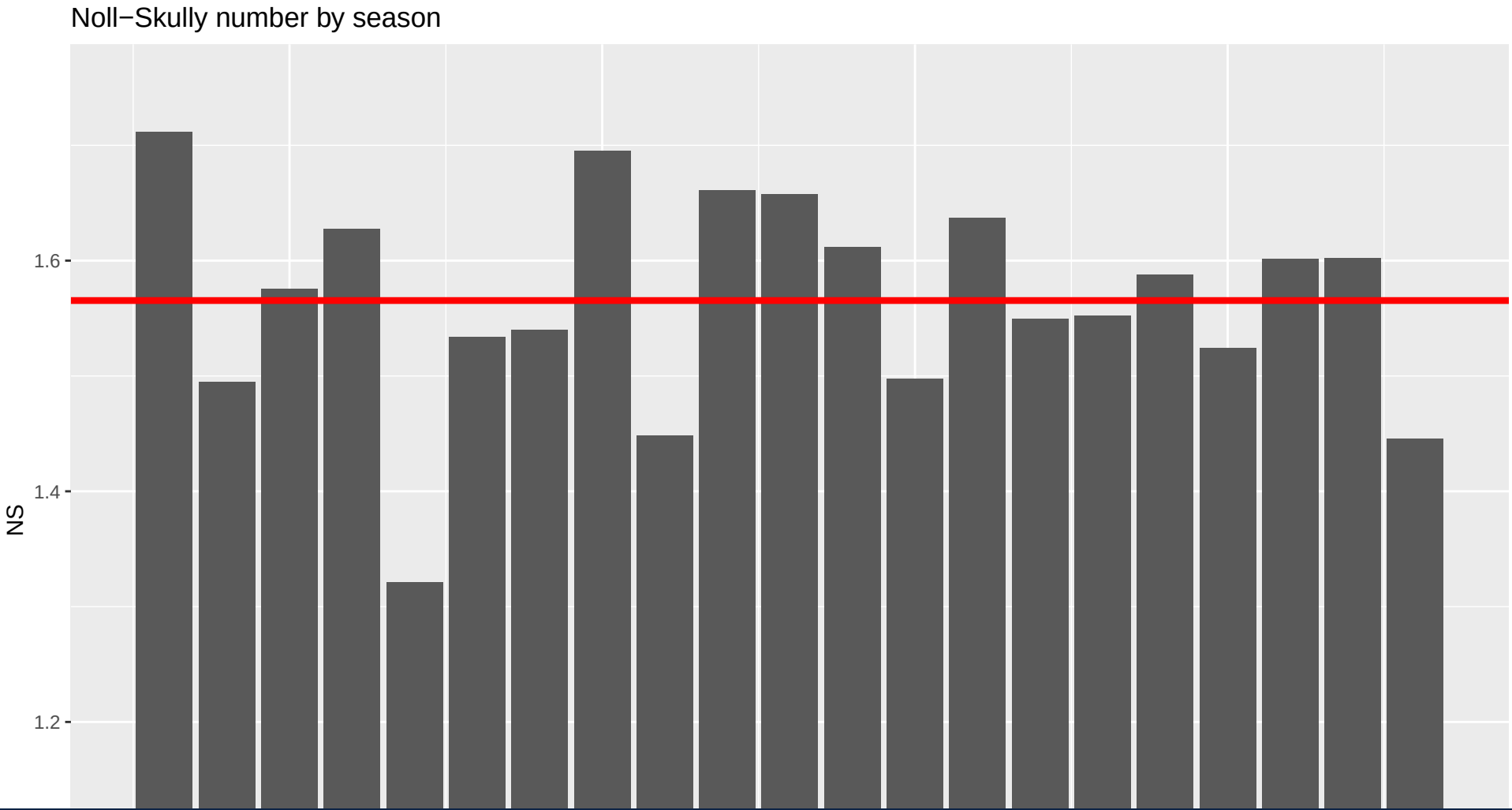
Plotting results

```
library(ggplot2)
ggplot(nfl_s, aes(Season, NS))+geom_bar(stat="identity")+
  coord_cartesian(ylim=c(1,1.75))+
  ggtitle("Noll-Skully number by season")
```



Measuring Seasonal Dimensions: Noll-Scully number

```
ggplot(nfl_s, aes(Season, NS))+geom_bar(stat="identity")+
  ggtitle("Noll-Skully number by season")+
  coord_cartesian(ylim=c(1,1.75))+
  geom_hline(yintercept=mean(nfl_s$NS), col="red", size=1.5)
```



Measuring Seasonal Dimensions: Noll-Scully number

CI for championship

How many times did every team become a champion?

```
champ_c<- seriea_st %>%
  filter(Rank ==1) %>%
  group_by(Team) %>%
  summarize(champ=n()) %>%
  arrange(desc(champ))

champ_c
```

```
## # A tibble: 5 x 2
##   Team        champ
##   <chr>       <int>
## 1 Juventus       15
## 2 AC Milan        4
## 3 Inter           4
## 4 AS Roma         1
## 5 Lazio Roma      1
```

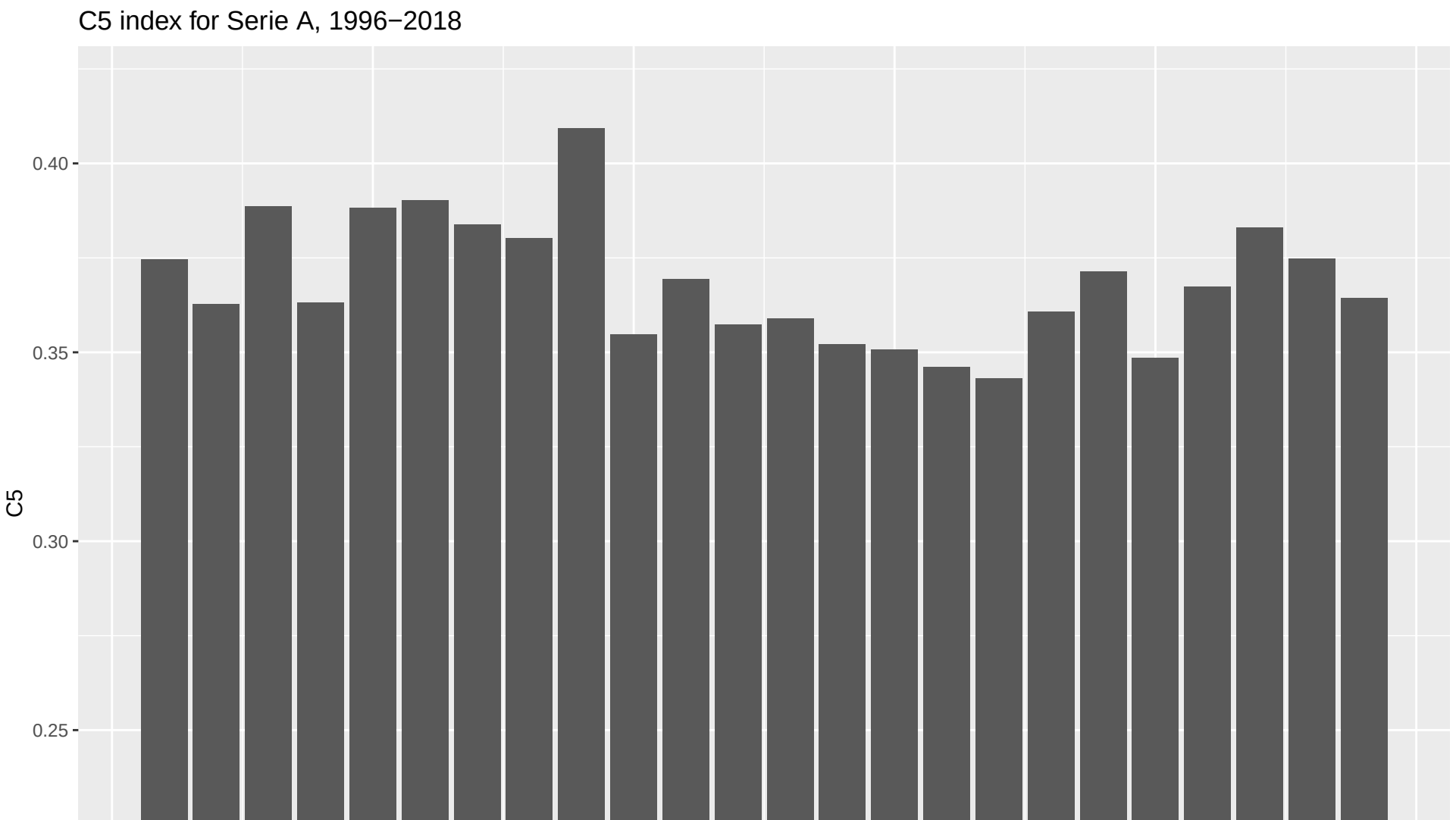
CI4 for championship

```
100*sum(champ_c[1:4,2])/sum(champ_c$champ)
```

```
## [1] 96
```

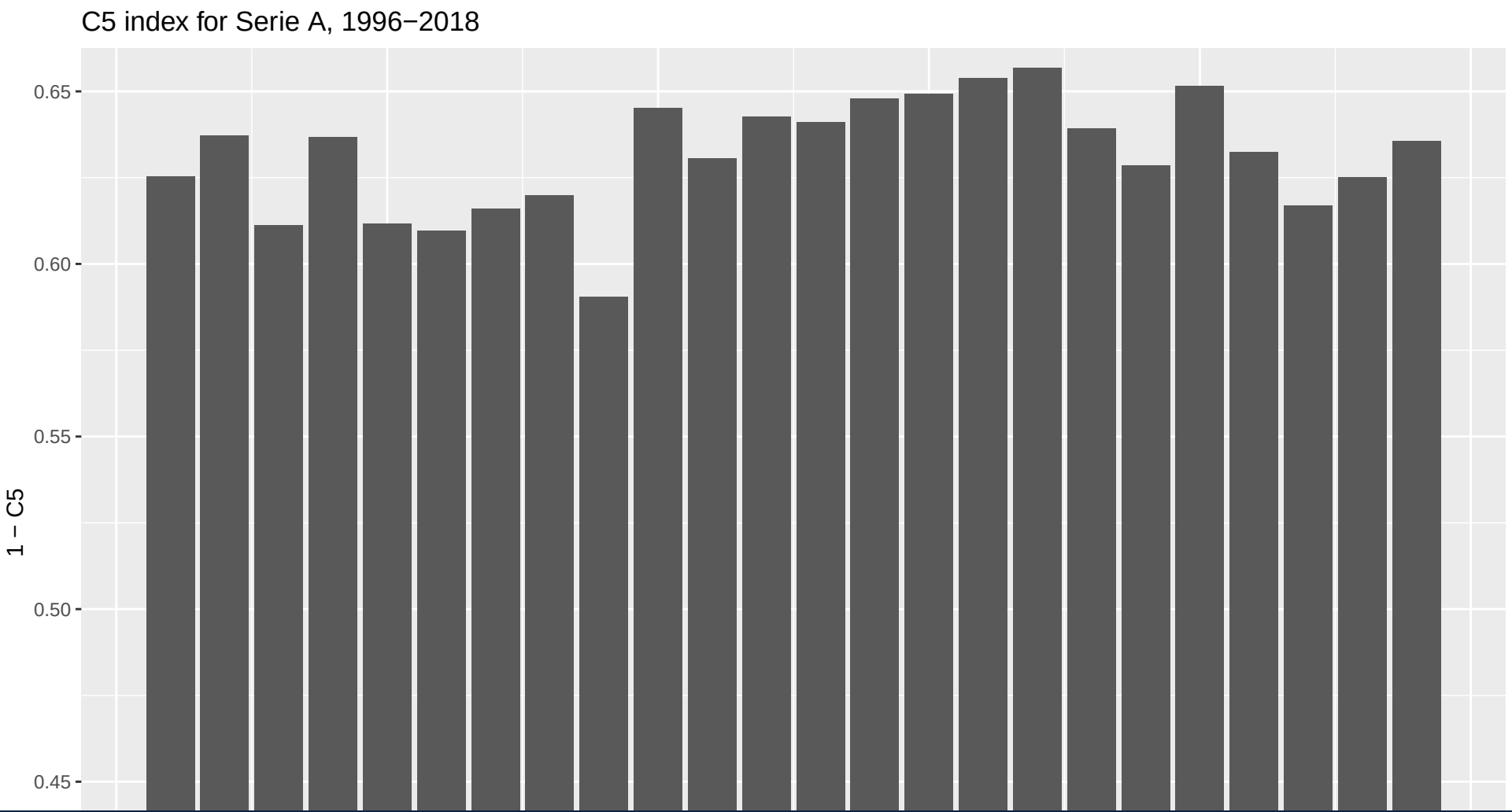
Measuring Seasonal Dimensions: Noll-Scully number

```
ggplot(C5, aes(Season, C5))+geom_bar(stat="identity")+
  ggtitle("C5 index for Serie A, 1996-2018")+
  coord_cartesian(ylim=c(0.2,0.42))
```



Measuring Seasonal Dimensions: Noll-Scully number

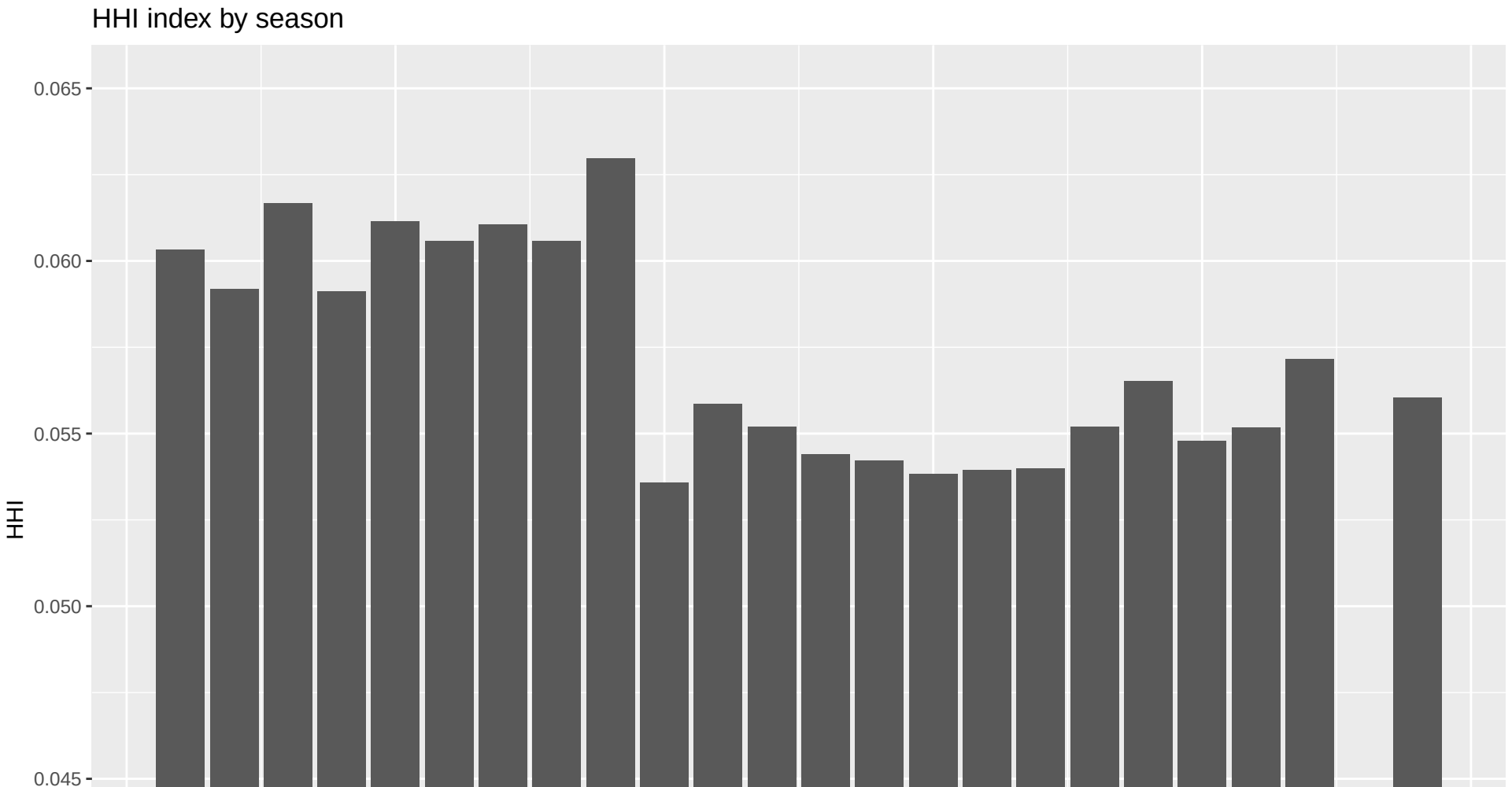
```
library(ggplot2)
ggplot(C5, aes(Season, 1-C5))+geom_bar(stat="identity")+
  ggtitle("C5 index for Serie A, 1996-2018")+
  coord_cartesian(ylim=c(0.4,0.65))
```



Measuring Seasonal Dimensions: Noll-Scully number

Plotting

```
ggplot(HHI, aes(Season, HHI))+geom_bar(stat="identity")+
  ggtitle("HHI index by season")+
  coord_cartesian(ylim=c(0.04,0.065))
```



Measuring Seasonal Dimensions: Noll-Scully number

```
library(SportsAnalytics270)

str(mlb_standings)
```

```
## 'data.frame':    600 obs. of  10 variables:
##  $ Season: int  1999 1999 1999 1999 1999 1999 1999 1999 1999 1999 ...
##  $ Lig   : chr  "AL" "NL" "NL" "AL" ...
##  $ Div   : chr  "W" "W" "E" "E" ...
##  $ Team  : chr  "Anaheim Angels" "Arizona Diamondbacks" "Atlanta Braves" "Baltimo
##  $ W     : int  70 100 103 78 94 75 67 96 97 72 ...
##  $ L     : int  92 62 59 84 68 86 95 67 65 90 ...
##  $ G     : int  162 162 162 162 162 162 162 163 162 162 ...
##  $ R     : int  711 908 840 851 836 777 747 865 1009 906 ...
##  $ RA    : int  826 676 661 815 718 870 920 711 860 1028 ...
##  $ Rank  : int  4 1 1 4 2 2 6 2 1 5 ...
```

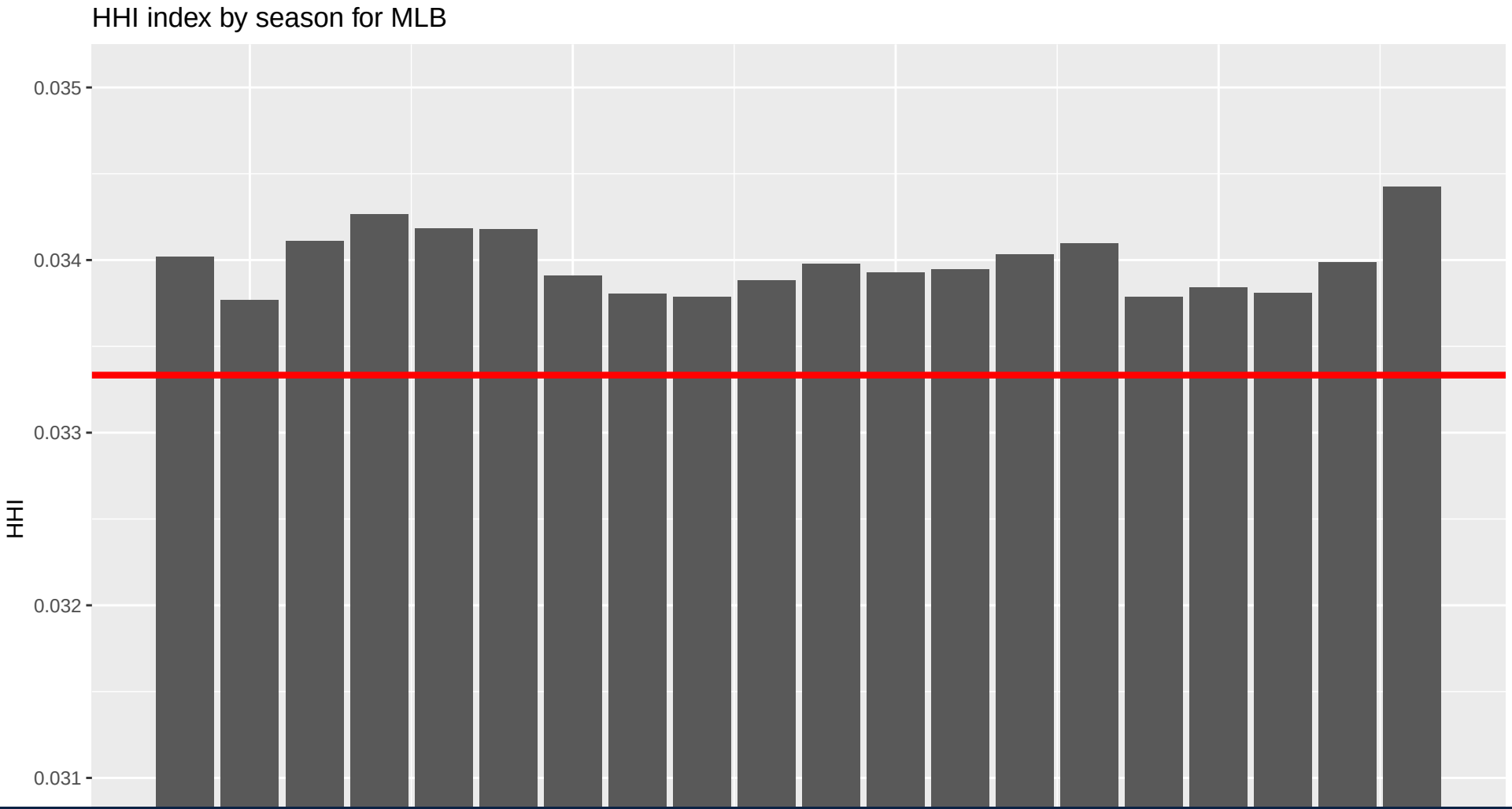
Measuring Seasonal Dimensions: Noll-Scully number

HHI Index

```
mlb_HHI <- mlb_standings %>%
      group_by(Season) %>%
      summarise(HHI=sum(W^2)*(4/(30^2*162^2)))
```

Measuring Seasonal Dimensions: Noll-Scully number

```
ggplot(mlb_HHI, aes(Season, HHI))+geom_bar(stat="identity")+
  geom_hline(yintercept = 1/30, col="red", size=1.5)+
  coord_cartesian(ylim=c(0.03,0.035))+
  ggtitle("HHI index by season for MLB")
```



Measuring Seasonal Dimensions: Noll-Scully number

Leading indicators: Salaries

```
library(SportsAnalytics270)
data(mlb_salary)
str(mlb_salary)
```

```
## 'data.frame':    638 obs. of  6 variables:
##  $ Team       : chr  "New York Yankees" "Texas Rangers" "Atlanta Braves" "Clevela
##  $ Opening.day: num  88130709 81301598 75065000 73857962 71720000 ...
##  $ Current    : num  88130709 81301598 75065000 73857962 71720000 ...
##  $ Avg        : num  3147525 2803503 2345781 2051610 2173333 ...
##  $ Median     : num  2008333 2225000 975000 670000 1500000 ...
##  $ Season     : num  1999 1999 1999 1999 1999 ...
```

Aggregate data by year

```
mlb_s<-mlb_salary %>%
      group_by(Season) %>%
      summarise(Tot.salary=sum(Current))
```

```
cor(mlb_s$Tot.salary, mlb_HHI$HHI)
```

```
## [1] -0.009126879
```

What about the standard deviation of the salary ?

```
mlb_s<-mlb_salary %>%
      group_by(Season) %>%
```

Measuring Seasonal Dimensions: Noll-Scully number

```
data("f_data_sm")
f_2009 <- f_data_sm %>%
  filter(SEASON > 2008,
         COUNTRY %in% c("Italy", "Germany", "Spain","England",
            "Netherlands", "Portugal"), !is.na(H) )
head(f_2009)
```

```
##   SEASON COUNTRY         LEAGUE       DATE      HOMETEAM  AWAYTEAM FTSC FTR
## 1   2009 England Premier League 2008-08-16       Arsenal West Brom  1-0   H
## 2   2009 England Premier League 2008-08-16        Bolton     Stoke  3-1   H
## 3   2009 England Premier League 2008-08-16       Everton Blackburn  2-3   A
## 4   2009 England Premier League 2008-08-16          Hull    Fulham  2-1   H
## 5   2009 England Premier League 2008-08-16 Middlesbrough Tottenham  2-1   H
## 6   2009 England Premier League 2008-08-16    Sunderland Liverpool  0-1   A
##   FTHG FTAG FTTG    H   D     A
## 1    1    0    1 1.20 6.5 15.00
## 2    3    1    4 1.83 3.5  4.50
## 3    2    3    5 2.00 3.3  4.00
## 4    2    1    3 2.60 3.2  2.80
## 5    2    1    3 3.20 3.4  2.25
## 6    0    1    1 5.50 3.6  1.67
```

Measuring Seasonal Dimensions: Noll-Scully number

```
library(odds.converter)
x <- odds.fv(c(1.4,3.8,8), input = 'dec', output = 'prob')
x
```

```
## [1] 0.6479113 0.2387042 0.1133845
```

```
sum(x)
```

```
## [1] 1
```

Measuring Seasonal Dimensions: Noll-Scully number

```
library(odds.converter)
f_2009[,c("H", "D", "A")] <- odds.fv(f_2009[,c("H", "D", "A")],
                        input = 'dec',output = 'prob')
head(f_2009)
```

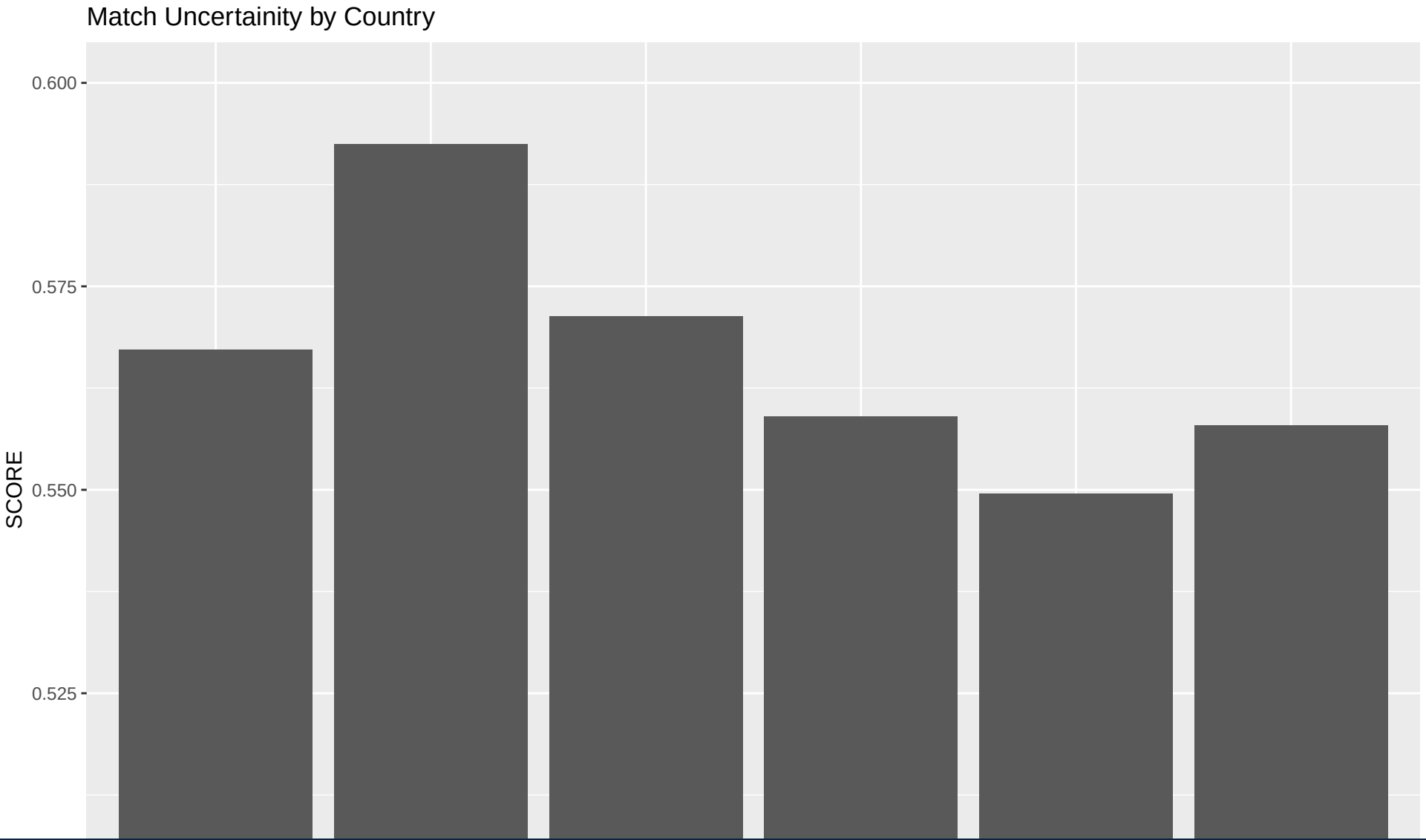
```
##   SEASON COUNTRY         LEAGUE       DATE      HOMETEAM  AWAYTEAM FTSC FTR
## 1   2009 England Premier League 2008-08-16       Arsenal West Brom  1-0   H
## 2   2009 England Premier League 2008-08-16        Bolton     Stoke  3-1   H
## 3   2009 England Premier League 2008-08-16       Everton Blackburn  2-3   A
## 4   2009 England Premier League 2008-08-16          Hull    Fulham  2-1   H
## 5   2009 England Premier League 2008-08-16 Middlesbrough Tottenham  2-1   H
## 6   2009 England Premier League 2008-08-16    Sunderland Liverpool  0-1   A
##   FTHG FTAG FTTG         H         D         A
## 1    1    0    1 0.7907543 0.1459854 0.06326034
## 2    3    1    4 0.5182626 0.2709773 0.21076012
## 3    2    3    5 0.4748201 0.2877698 0.23741007
## 4    2    1    3 0.3648208 0.2964169 0.33876221
## 5    2    1    3 0.2973183 0.2798290 0.42285270
## 6    0    1    1 0.1717862 0.2624511 0.56576278
```

```
library(measures)

brier_score <- f_2009 %>% group_by(COUNTRY) %>%
  do(SCORE = multiclass.Brier(.[,c("H", "D", "A")], .$FTR))
brier_score <- as.data.frame(brier_score)
```

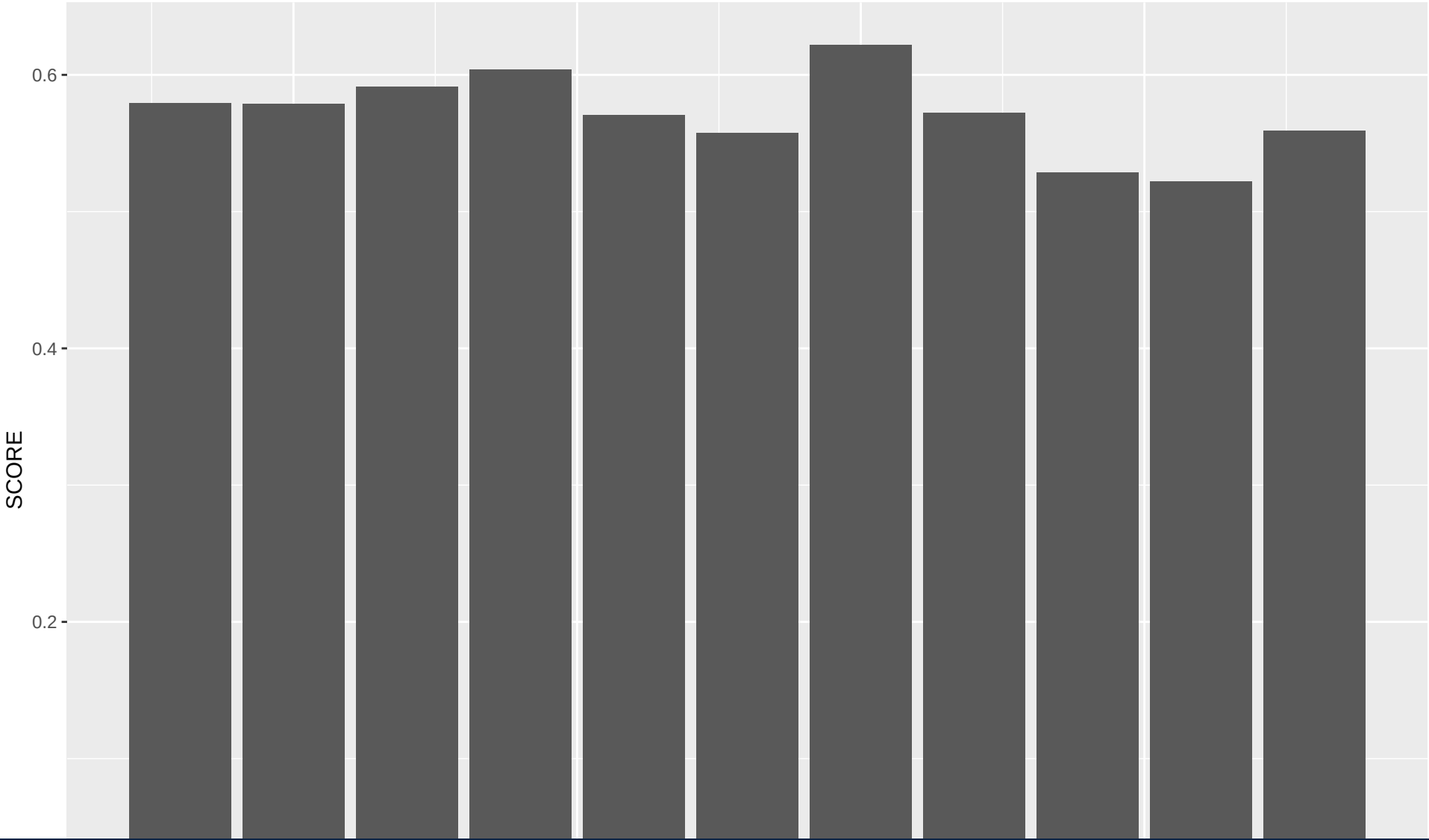
Measuring Seasonal Dimensions: Noll-Scully number

```
ggplot(brier_score, aes(x=COUNTRY, y = SCORE))+geom_bar(stat='identity')+
  coord_cartesian(ylim = c(0.5,0.6))+ggtitle('Match Uncertainity by Country')
```



Measuring Seasonal Dimensions: Noll-Scully number

```
brier_score %>% filter(COUNTRY=='Italy') %>%
ggplot(aes(x=SEASON, y=SCORE))+geom_bar(stat='identity')
```



Measuring Seasonal Dimensions: Noll-Scully number

```
ggplot(brier_score, aes(x=SEASON, y=SCORE))+
  geom_bar(stat='identity',position=position_dodge(width=1))+
  facet_grid(~COUNTRY)+ coord_flip()+ggtitle("Brier Score by Country")
```



Noll-Scully number: Calculations

```
library(SportsAnalytics270)
data("nfl_st")
str(nfl_st)
```

```
## 'data.frame':    667 obs. of  16 variables:
##  $ Tm    : chr  "New York Jets" "Miami Dolphins" "Buffalo Bills" "New England Pat
##  $ W     : num  12 10 10 9 3 11 8 7 6 3 ...
##  $ L     : num  4 6 6 7 13 5 8 9 10 13 ...
##  $ T     : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ Wpct  : num  0.75 0.625 0.625 0.563 0.188 0.688 0.5 0.438 0.375 0.188 ...
##  $ PF    : num  416 321 400 337 310 392 330 263 269 268 ...
##  $ PA    : num  266 265 333 329 444 338 320 303 335 452 ...
##  $ PD    : num  150 56 67 8 -134 54 10 -40 -66 -184 ...
##  $ MoV   : num  9.4 3.5 4.2 0.5 -8.4 3.4 0.6 -2.5 -4.1 -11.5 ...
##  $ SoS   : num  1.8 2.4 1 3.4 3 -0.9 -0.6 -1.3 -1.3 0.5 ...
##  $ SRS   : num  11.2 5.9 5.2 3.9 -5.4 2.5 0 -3.8 -5.4 -11 ...
##  $ OSRS  : num  5.4 -0.7 4.4 1.1 -0.1 3.1 -0.5 -5.1 -5.2 -4.5 ...
##  $ DSRS  : num  5.8 6.5 0.7 2.8 -5.3 -0.6 0.5 1.3 -0.2 -6.5 ...
##  $ Season: int  1998 1998 1998 1998 1998 1998 1998 1998 1998 1998 ...
##  $ Conf  : Factor w/ 2 levels "AFC","NFC": 1 1 1 1 1 1 1 1 1 1 ...
##  $ Div   : Factor w/ 5 levels "Central","East",..: 2 2 2 2 2 1 1 1 1 1 ...
```

Standard Deviations: Serie A

Noll-Scully number

```
#Ideal StDev
id_s<-0.5/sqrt(16)
# Win percentage StDev
sd_wpct<-sd(nfl_st$Wpct)
# Noll-Scully number
sd_wpct/id_s
```

```
## [1] 1.543827
```

Standard Deviations: Serie A

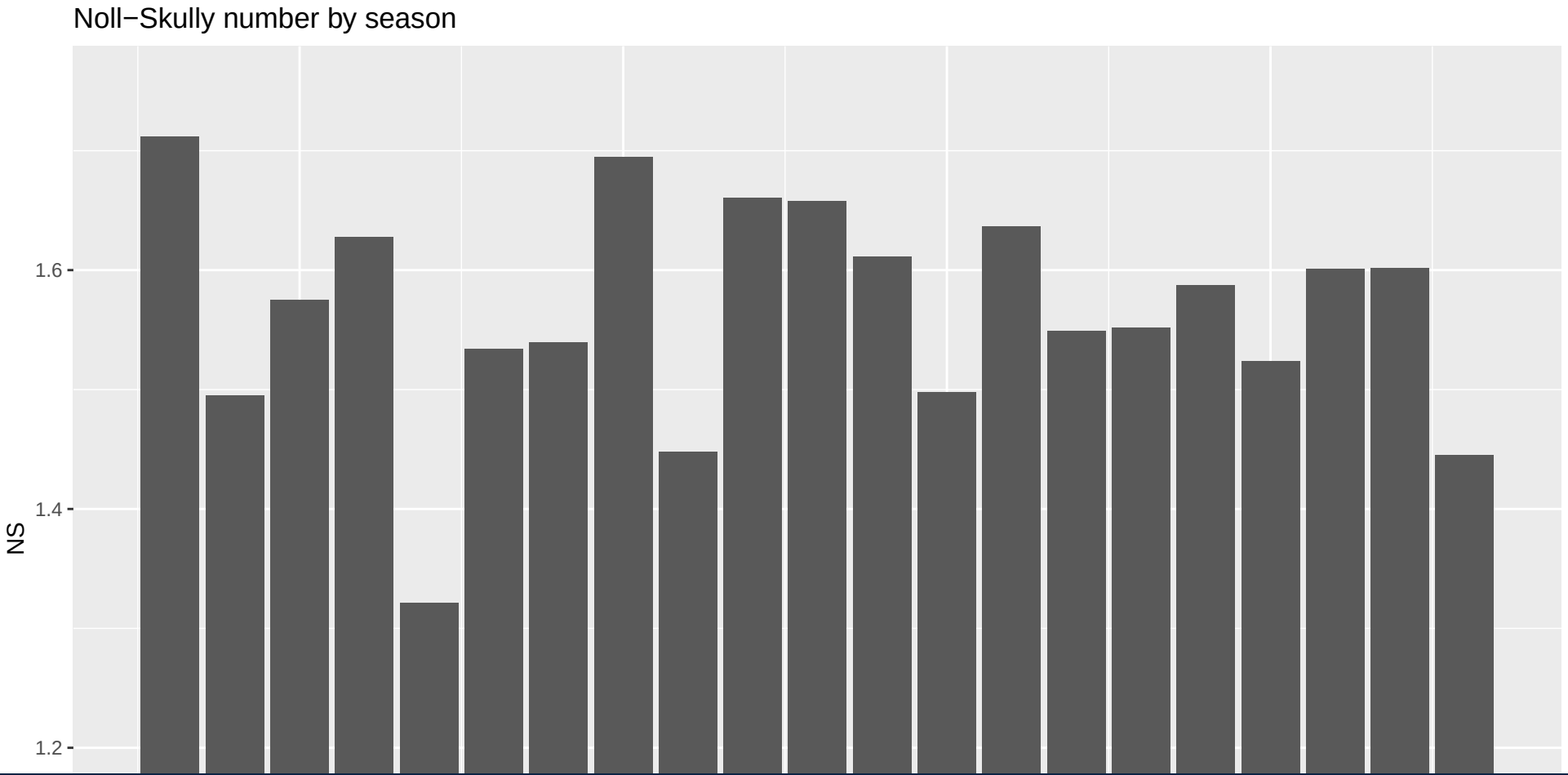
```
library(dplyr)
nfl_s<-nfl_st %>%
  group_by(Season) %>%
  summarise(NS=sd(Wpct)/id_s)
nfl_s
```

```
## # A tibble: 21 x 2
##    Season    NS
##     <int> <dbl>
##  1   1998  1.71
##  2   1999  1.49
##  3   2000  1.58
##  4   2001  1.63
##  5   2002  1.32
##  6   2003  1.53
##  7   2004  1.54
##  8   2005  1.70
##  9   2006  1.45
## 10   2007  1.66
## # ... with 11 more rows
```

Standard Deviations: Serie A

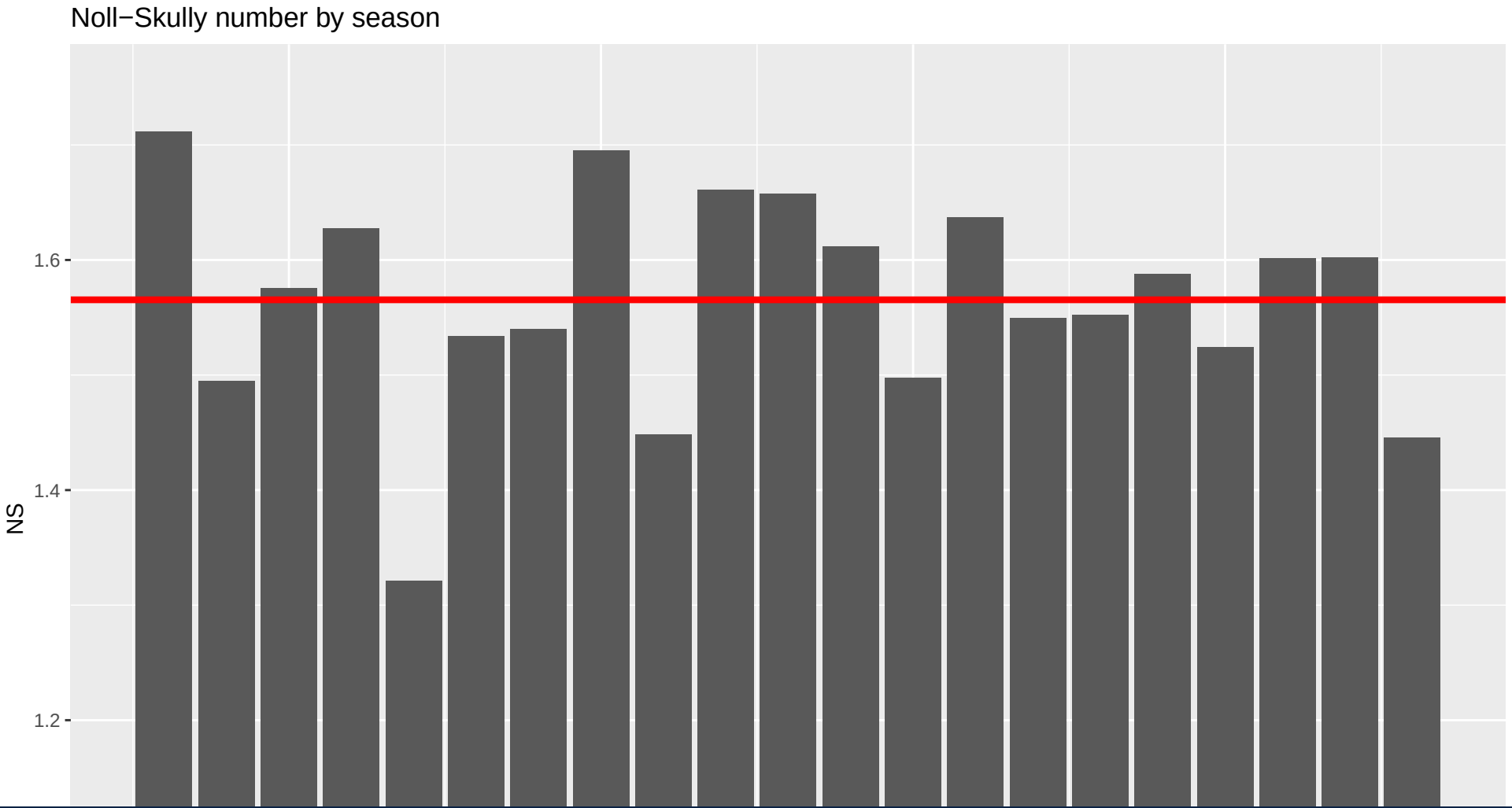
ploting

```
library(ggplot2)
ggplot(nfl_s, aes(Season, NS))+geom_bar(stat="identity")+
  coord_cartesian(ylim=c(1,1.75))+
  ggtitle("Noll-Skully number by season")
```



Standard Deviations: Serie A

```
ggplot(nfl_s, aes(Season, NS))+geom_bar(stat="identity")+
  ggtitle("Noll-Skully number by season")+
  coord_cartesian(ylim=c(1,1.75))+
  geom_hline(yintercept=mean(nfl_s$NS), col="red", size=1.5)
```



Standard Deviations: Serie A

CI for championship

How many times did every team become a champion?

```
champ_c<- seriea_st %>%
  filter(Rank ==1) %>%
  group_by(Team) %>%
  summarize(champ=n()) %>%
  arrange(desc(champ))

champ_c
```

```
## # A tibble: 5 x 2
##   Team        champ
##   <chr>       <int>
## 1 Juventus       15
## 2 AC Milan        4
## 3 Inter           4
## 4 AS Roma         1
## 5 Lazio Roma      1
```

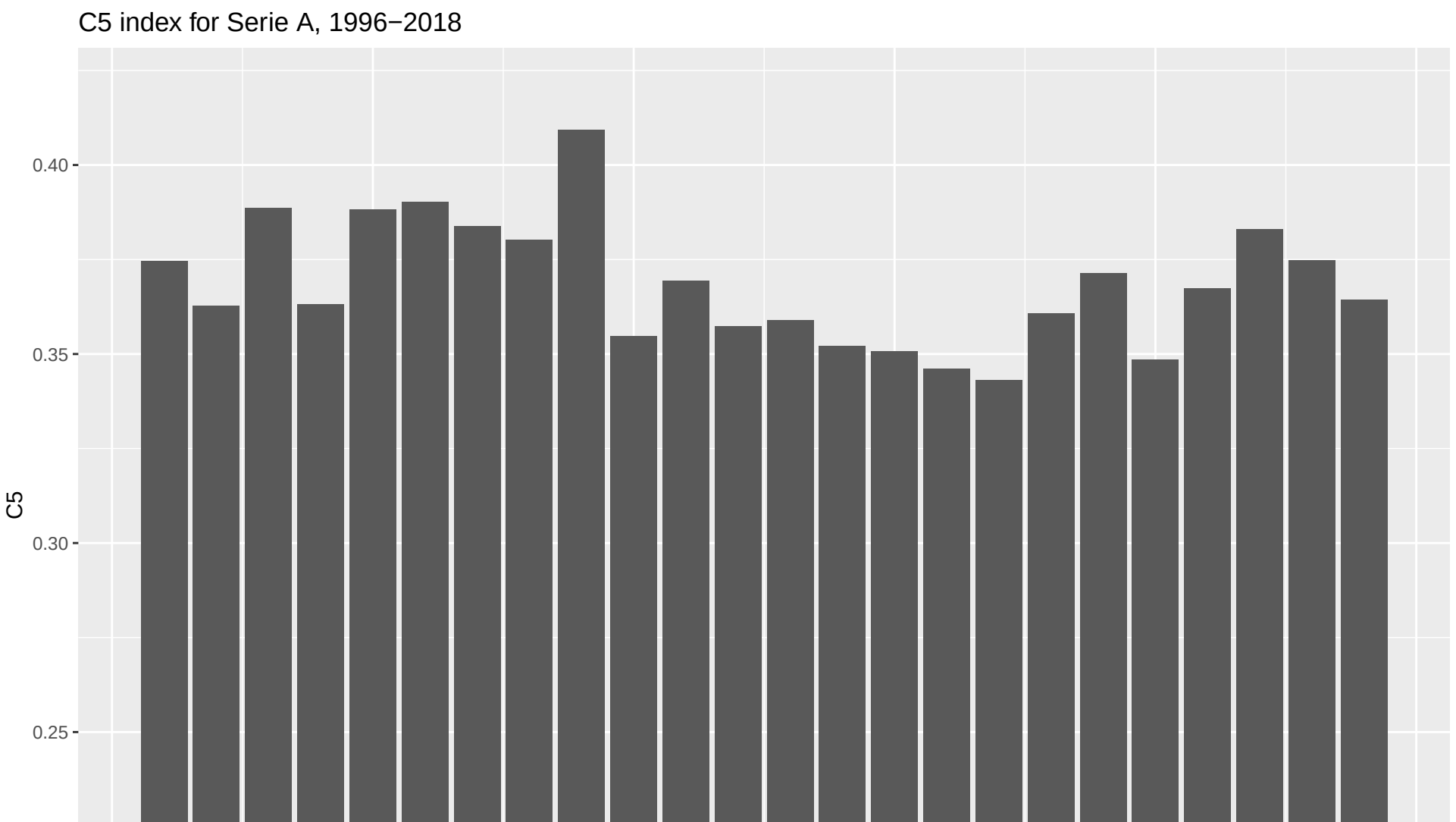
CI4 for championship

```
100*sum(champ_c[1:4,2])/sum(champ_c$champ)
```

```
## [1] 96
```

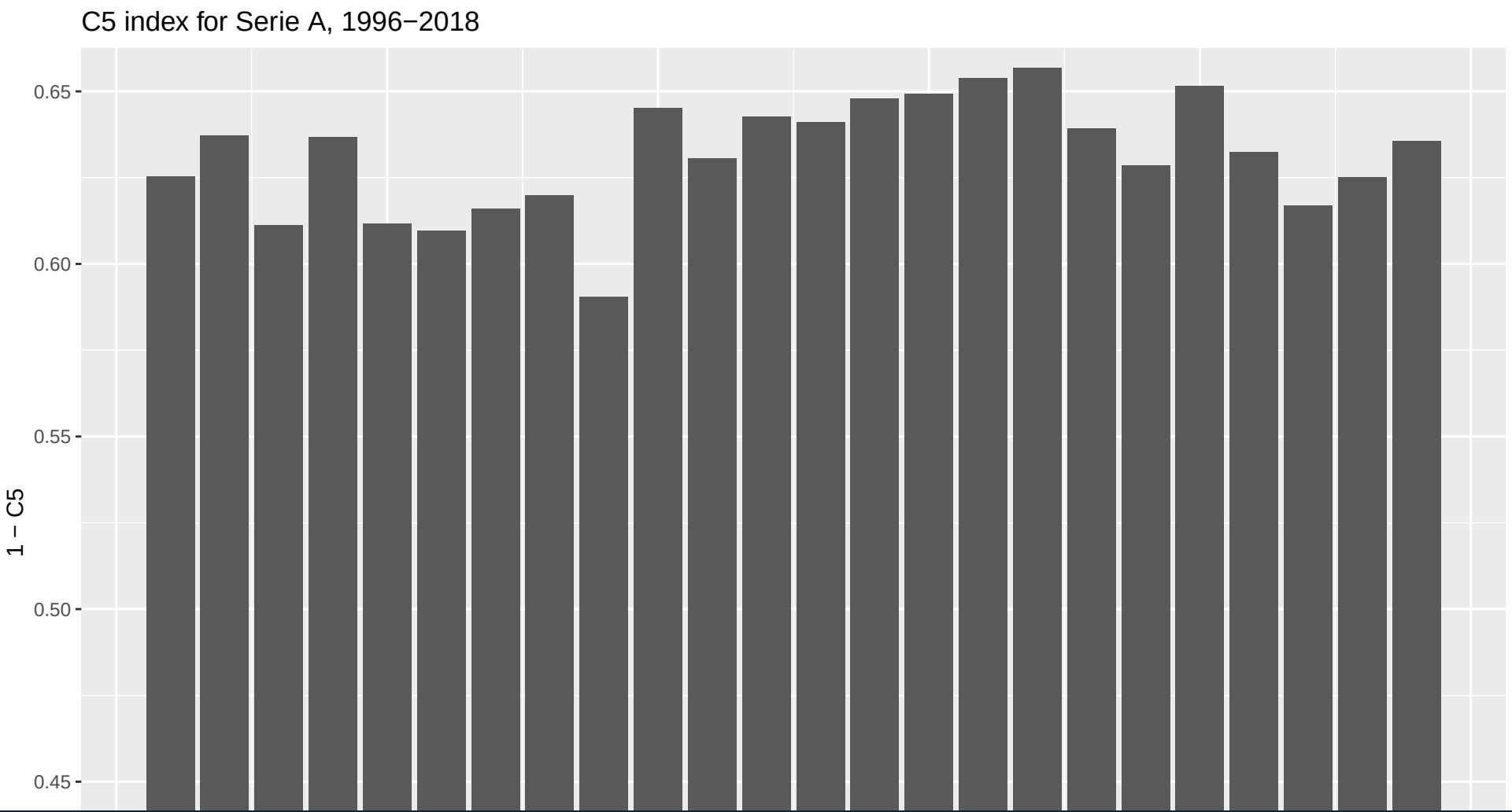
Standard Deviations: Serie A

```
ggplot(C5, aes(Season, C5))+geom_bar(stat="identity")+
  ggtitle("C5 index for Serie A, 1996-2018")+
  coord_cartesian(ylim=c(0.2,0.42))
```



Standard Deviations: Serie A

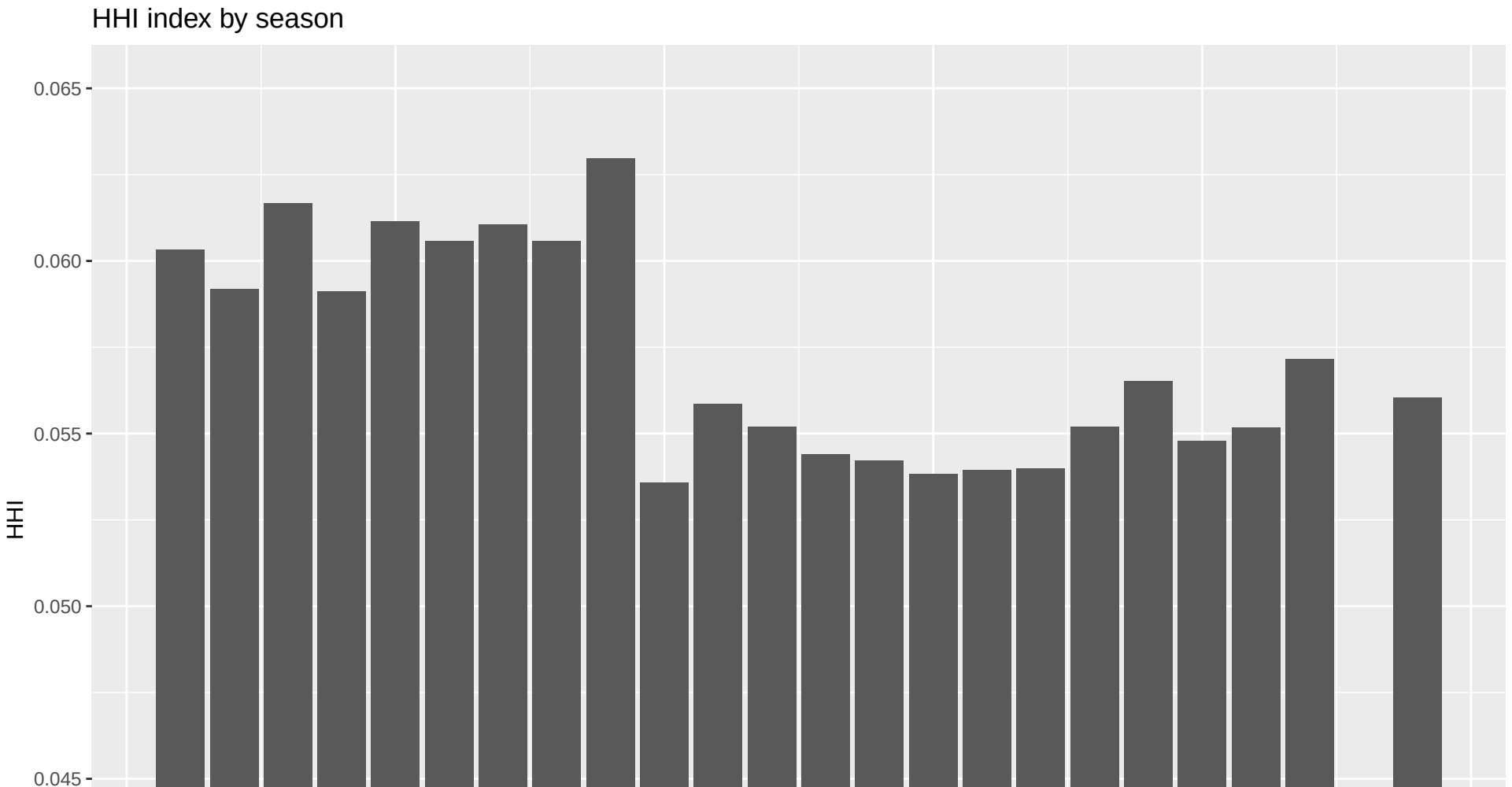
```
library(ggplot2)
ggplot(C5, aes(Season, 1-C5))+geom_bar(stat="identity")+
  ggtitle("C5 index for Serie A, 1996-2018")+
  coord_cartesian(ylim=c(0.4,0.65))
```



Standard Deviations: Serie A

Plotting

```
ggplot(HHI, aes(Season, HHI))+geom_bar(stat="identity")+
  ggtitle("HHI index by season")+
  coord_cartesian(ylim=c(0.04,0.065))
```



Standard Deviations: Serie A

```
library(SportsAnalytics270)

str(mlb_standings)
```

```
## 'data.frame':    600 obs. of  10 variables:
##  $ Season: int  1999 1999 1999 1999 1999 1999 1999 1999 1999 1999 ...
##  $ Lig   : chr  "AL" "NL" "NL" "AL" ...
##  $ Div   : chr  "W" "W" "E" "E" ...
##  $ Team  : chr  "Anaheim Angels" "Arizona Diamondbacks" "Atlanta Braves" "Baltimo
##  $ W     : int  70 100 103 78 94 75 67 96 97 72 ...
##  $ L     : int  92 62 59 84 68 86 95 67 65 90 ...
##  $ G     : int  162 162 162 162 162 162 162 163 162 162 ...
##  $ R     : int  711 908 840 851 836 777 747 865 1009 906 ...
##  $ RA    : int  826 676 661 815 718 870 920 711 860 1028 ...
##  $ Rank  : int  4 1 1 4 2 2 6 2 1 5 ...
```

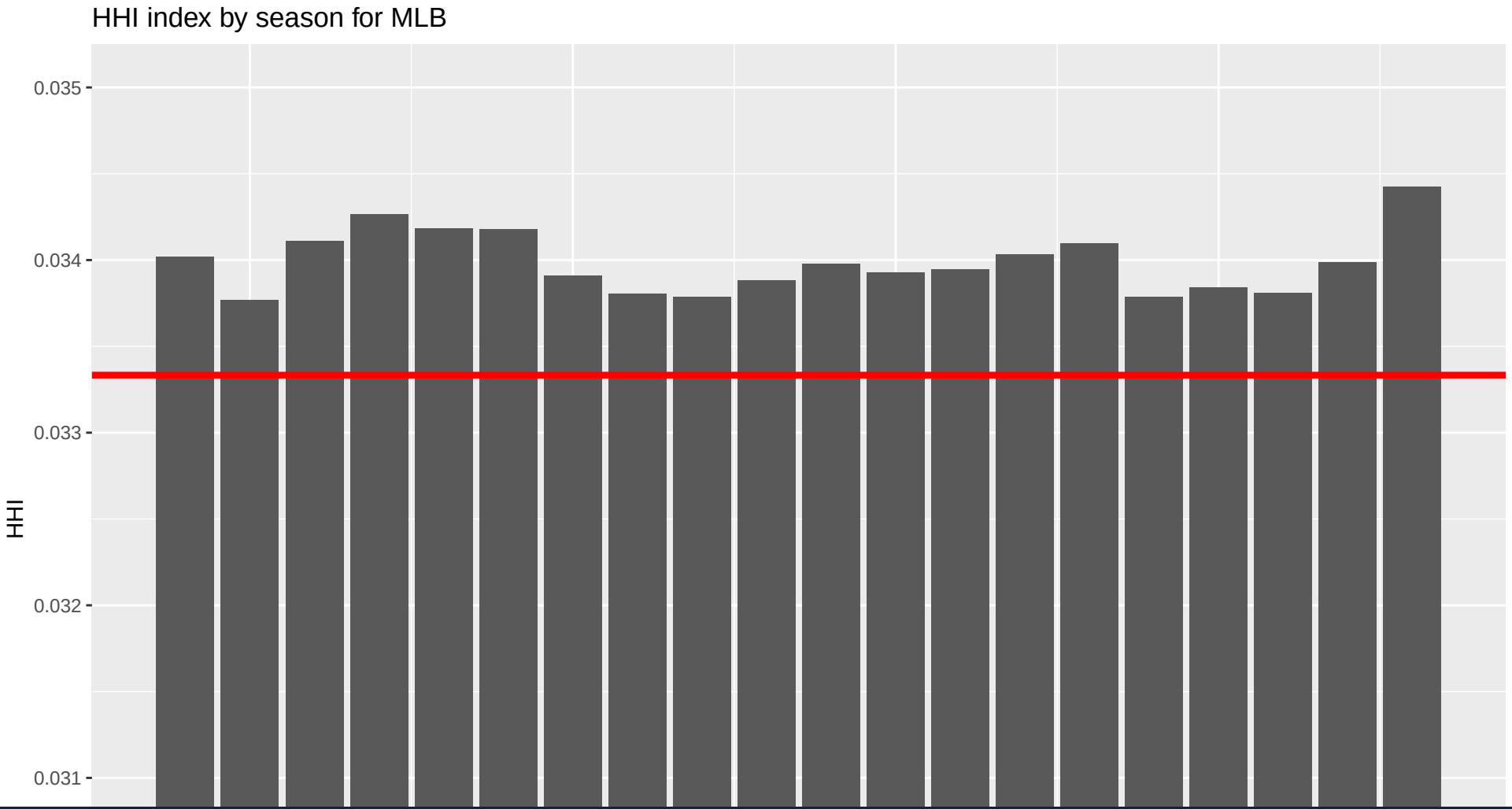
Standard Deviations: Serie A

HHI Index

```
mlb_HHI <- mlb_standings %>%
      group_by(Season) %>%
      summarise(HHI=sum(W^2)*(4/(30^2*162^2)))
```

Standard Deviations: Serie A

```
ggplot(mlb_HHI, aes(Season, HHI))+geom_bar(stat="identity")+
  geom_hline(yintercept = 1/30, col="red", size=1.5)+
  coord_cartesian(ylim=c(0.03,0.035))+
  ggtitle("HHI index by season for MLB")
```



Standard Deviations: Serie A

Leading indicators: Salaries

```
library(SportsAnalytics270)
data(mlb_salary)
str(mlb_salary)
```

```
## 'data.frame':    638 obs. of  6 variables:
##  $ Team       : chr  "New York Yankees" "Texas Rangers" "Atlanta Braves" "Clevela
##  $ Opening.day: num  88130709 81301598 75065000 73857962 71720000 ...
##  $ Current    : num  88130709 81301598 75065000 73857962 71720000 ...
##  $ Avg        : num  3147525 2803503 2345781 2051610 2173333 ...
##  $ Median     : num  2008333 2225000 975000 670000 1500000 ...
##  $ Season     : num  1999 1999 1999 1999 1999 ...
```

Aggregate data by year

```
mlb_s<-mlb_salary %>%
      group_by(Season) %>%
      summarise(Tot.salary=sum(Current))
```

```
cor(mlb_s$Tot.salary, mlb_HHI$HHI)
```

```
## [1] -0.009126879
```

What about the standard deviation of the salary ?

```
mlb_s<-mlb_salary %>%
      group_by(Season) %>%
```

Standard Deviations: Serie A

```
data("f_data_sm")
f_2009 <- f_data_sm %>%
  filter(SEASON > 2008,
         COUNTRY %in% c("Italy", "Germany", "Spain","England",
            "Netherlands", "Portugal"), !is.na(H) )
head(f_2009)
```

```
##   SEASON COUNTRY         LEAGUE       DATE      HOMETEAM  AWAYTEAM FTSC FTR
## 1   2009 England Premier League 2008-08-16       Arsenal West Brom  1-0   H
## 2   2009 England Premier League 2008-08-16        Bolton     Stoke  3-1   H
## 3   2009 England Premier League 2008-08-16       Everton Blackburn  2-3   A
## 4   2009 England Premier League 2008-08-16          Hull    Fulham  2-1   H
## 5   2009 England Premier League 2008-08-16 Middlesbrough Tottenham  2-1   H
## 6   2009 England Premier League 2008-08-16    Sunderland Liverpool  0-1   A
##   FTHG FTAG FTTG    H   D     A
## 1    1    0    1 1.20 6.5 15.00
## 2    3    1    4 1.83 3.5  4.50
## 3    2    3    5 2.00 3.3  4.00
## 4    2    1    3 2.60 3.2  2.80
## 5    2    1    3 3.20 3.4  2.25
## 6    0    1    1 5.50 3.6  1.67
```

Standard Deviations: Serie A

```
library(odds.converter)
x <- odds.fv(c(1.4,3.8,8), input = 'dec', output = 'prob')
x
```

```
## [1] 0.6479113 0.2387042 0.1133845
```

```
sum(x)
```

```
## [1] 1
```

Standard Deviations: Serie A

```
library(odds.converter)
f_2009[,c("H", "D", "A")] <- odds.fv(f_2009[,c("H", "D", "A")],
                        input = 'dec',output = 'prob')
head(f_2009)
```

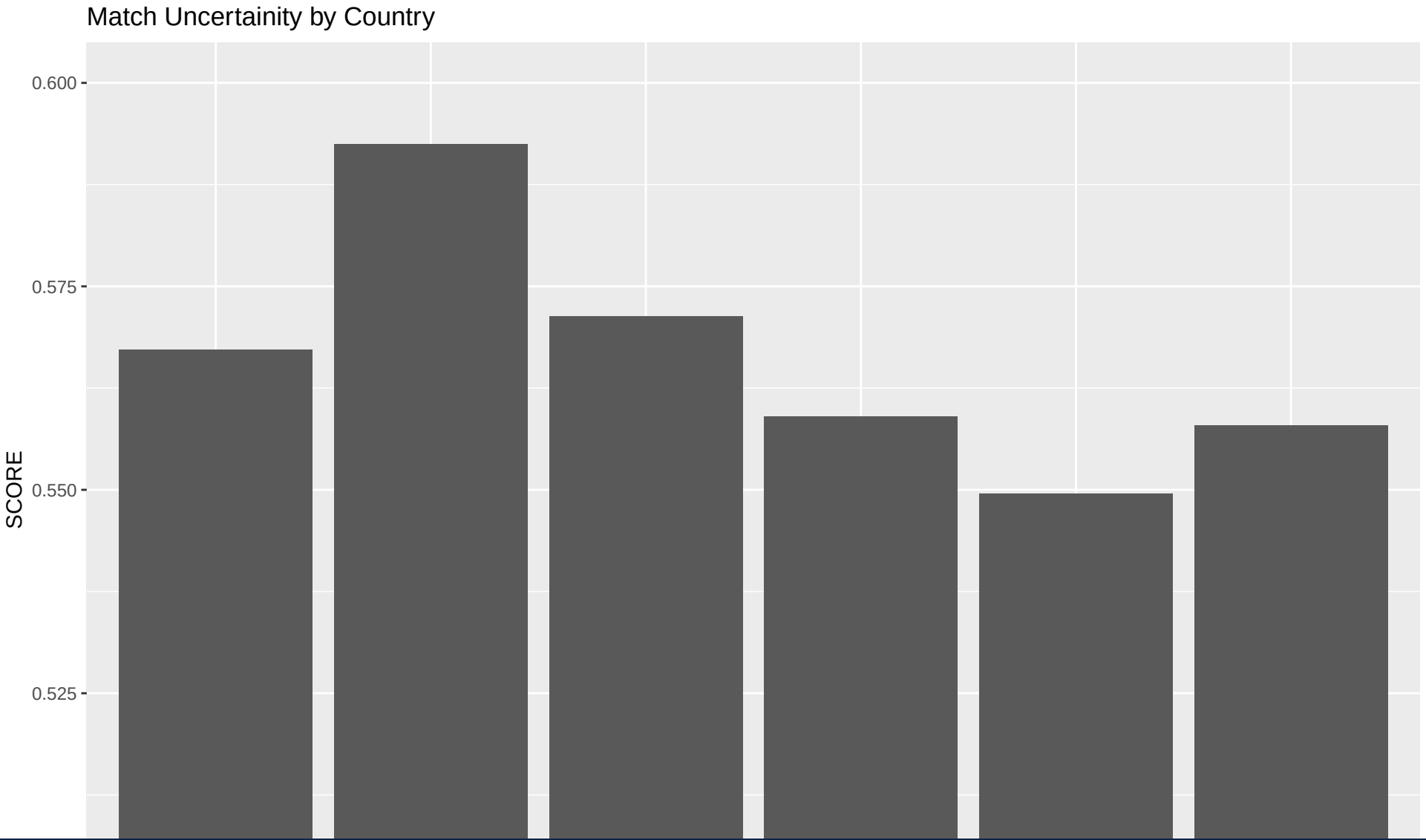
```
##   SEASON COUNTRY         LEAGUE       DATE      HOMETEAM  AWAYTEAM FTSC FTR
## 1   2009 England Premier League 2008-08-16       Arsenal West Brom  1-0   H
## 2   2009 England Premier League 2008-08-16        Bolton     Stoke  3-1   H
## 3   2009 England Premier League 2008-08-16       Everton Blackburn  2-3   A
## 4   2009 England Premier League 2008-08-16          Hull    Fulham  2-1   H
## 5   2009 England Premier League 2008-08-16 Middlesbrough Tottenham  2-1   H
## 6   2009 England Premier League 2008-08-16    Sunderland Liverpool  0-1   A
##   FTHG FTAG FTTG         H         D         A
## 1    1    0    1 0.7907543 0.1459854 0.06326034
## 2    3    1    4 0.5182626 0.2709773 0.21076012
## 3    2    3    5 0.4748201 0.2877698 0.23741007
## 4    2    1    3 0.3648208 0.2964169 0.33876221
## 5    2    1    3 0.2973183 0.2798290 0.42285270
## 6    0    1    1 0.1717862 0.2624511 0.56576278
```

```
library(measures)

brier_score <- f_2009 %>% group_by(COUNTRY) %>%
  do(SCORE = multiclass.Brier(.[,c("H", "D", "A")], .$FTR))
brier_score <- as.data.frame(brier_score)
```

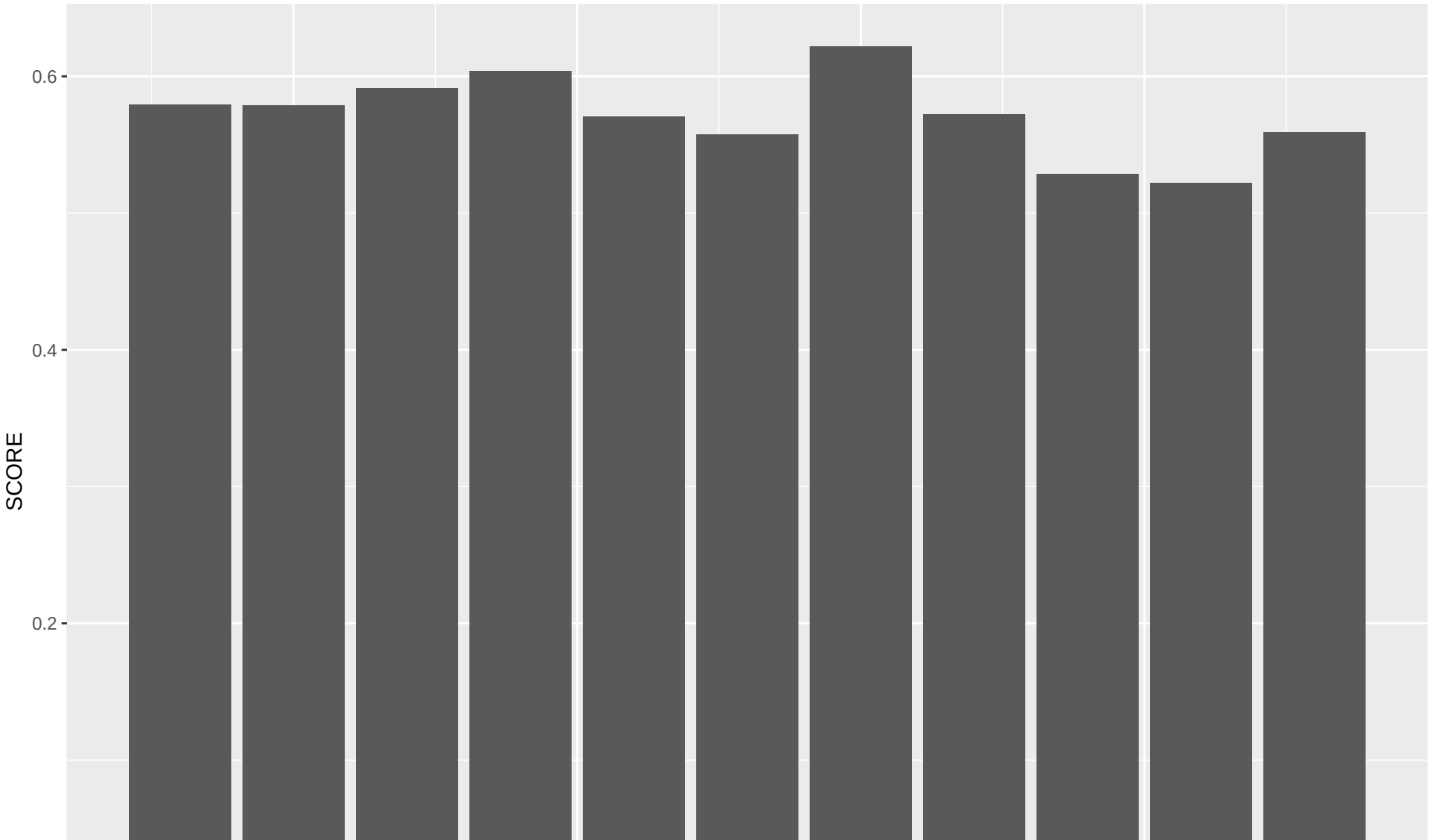
Standard Deviations: Serie A

```
ggplot(brier_score, aes(x=COUNTRY, y = SCORE))+geom_bar(stat='identity')+
  coord_cartesian(ylim = c(0.5,0.6))+ggtitle('Match Uncertainity by Country')
```



Standard Deviations: Serie A

```
brier_score %>% filter(COUNTRY=='Italy') %>%
ggplot(aes(x=SEASON, y=SCORE))+geom_bar(stat='identity')
```



Standard Deviations: Serie A

```
ggplot(brier_score, aes(x=SEASON, y=SCORE))+
  geom_bar(stat='identity',position=position_dodge(width=1))+
  facet_grid(~COUNTRY)+ coord_flip()+ggtitle("Brier Score by Country")
```

